

Backpropagation: Understanding How to Update ANNs Weights Step-by-Step

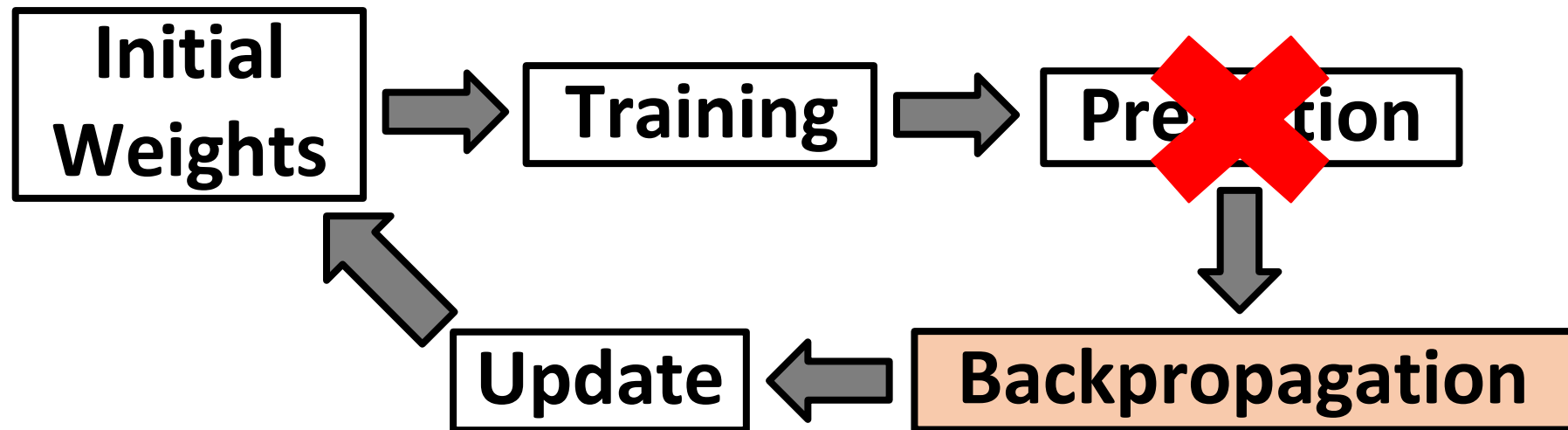
Train then Update

- The backpropagation algorithm is used to update the NN weights when they are not able to make the correct predictions. Hence, we should train the NN before applying backpropagation.

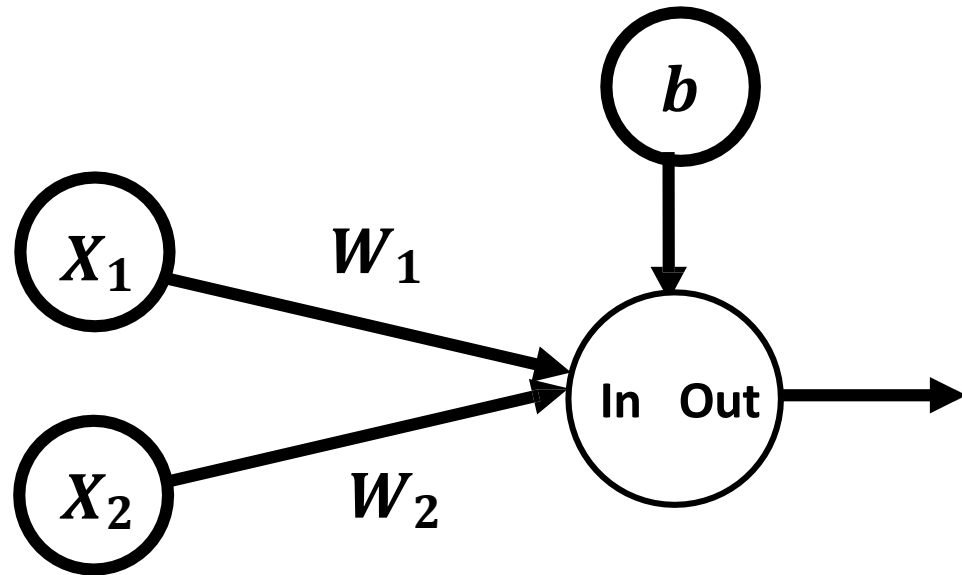


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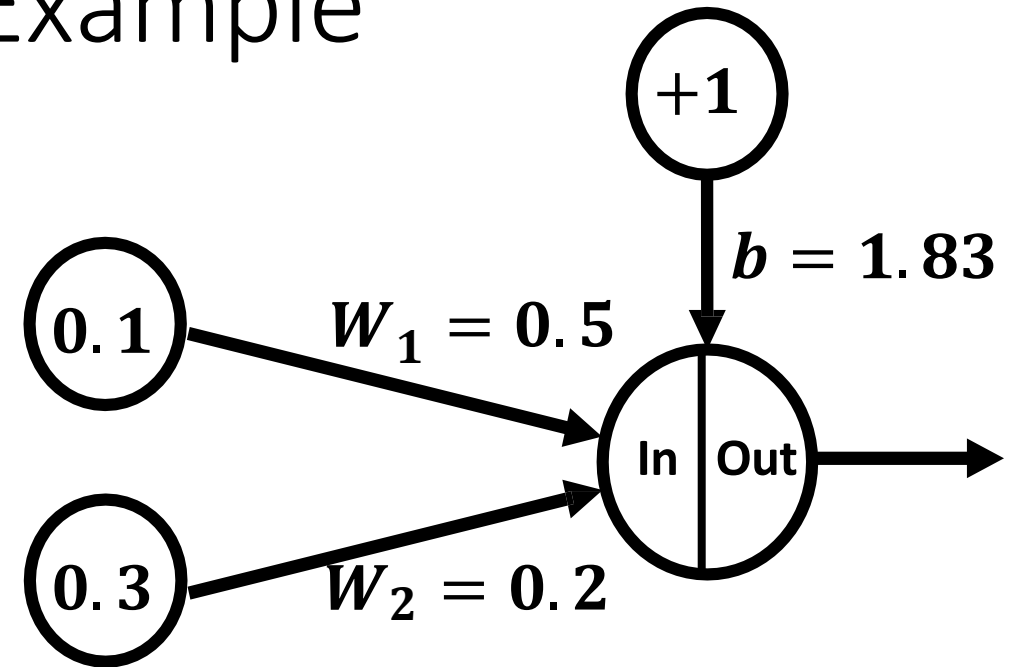


Neural Network Training Example



Training Data

X_1	X_2	Output
0.1	0.3	0.03

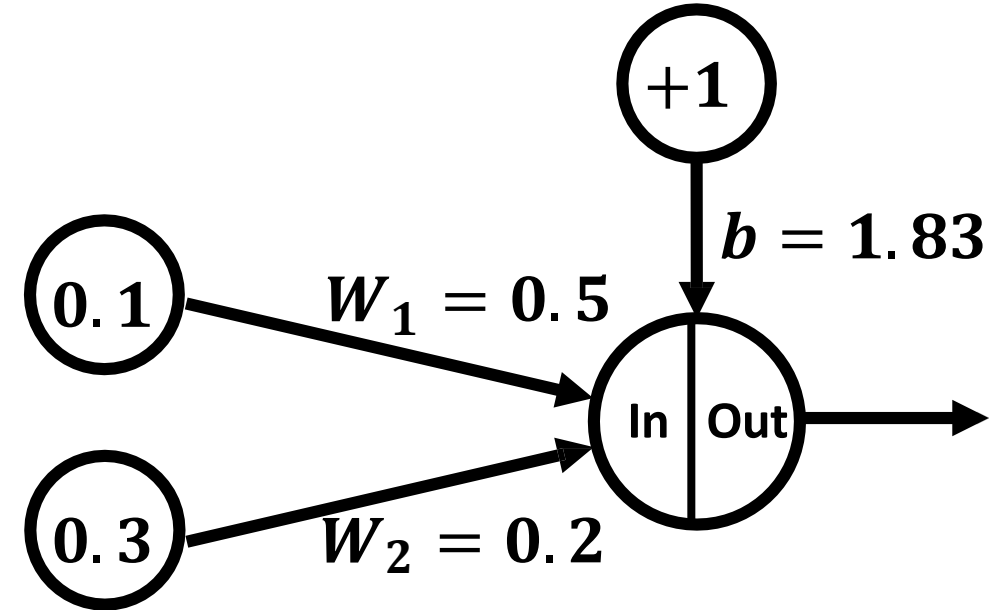


Initial Weights

W_1	W_2	b
0.5	0.5	1.83

Network Training

- Steps to train our network:
 1. Prepare activation function input (sum of products between inputs and weights).
 2. Activation function output.



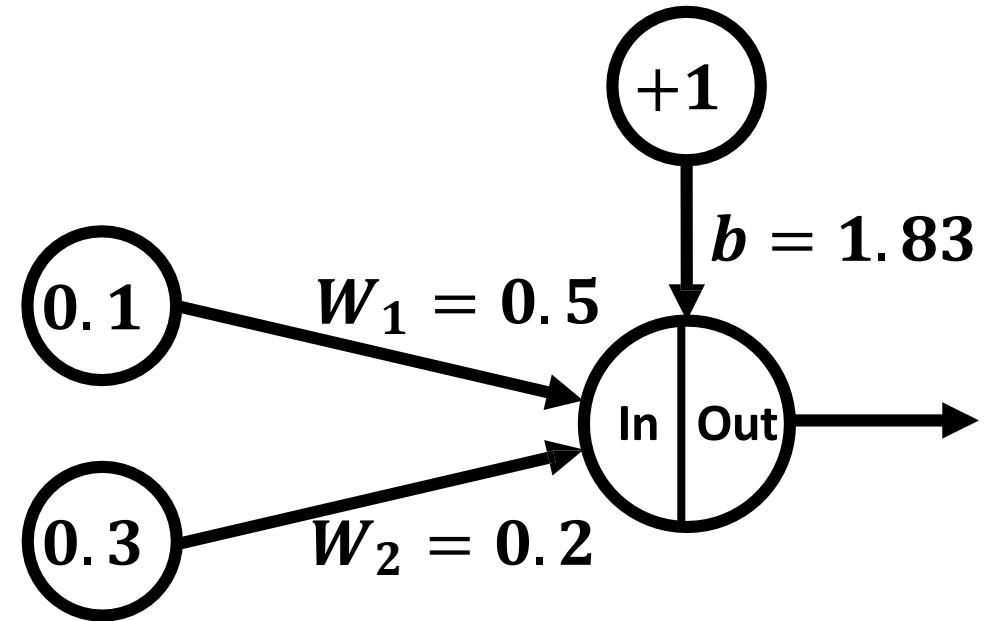
Network Training: Sum of Products

$$s = X_1 * W_1 + X_2 * W_2 + b$$

$$s = 0.1 * 0.5 + 0.3 * 0.2 + 1.83$$

$$s = 1.94$$

- After calculating the sop between inputs and weights, next is to use this sop as the input to the activation function.



Network Training: Activation Function

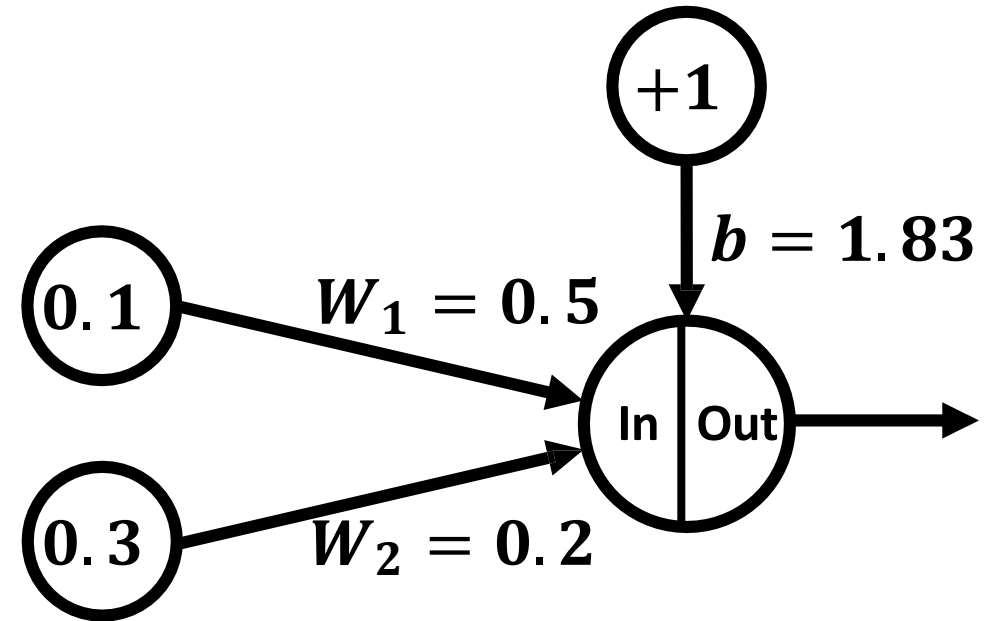
- In this example, the sigmoid activation function is used.

$$f(s) = \frac{1}{1 + e^{-s}}$$

- Based on the sop calculated previously, the output is as follows:

$$f(s) = \frac{1}{1 + e^{-1.94}} = \frac{1}{1 + 0.144} = \frac{1}{1.144}$$

$$f(s) = \mathbf{0.874}$$



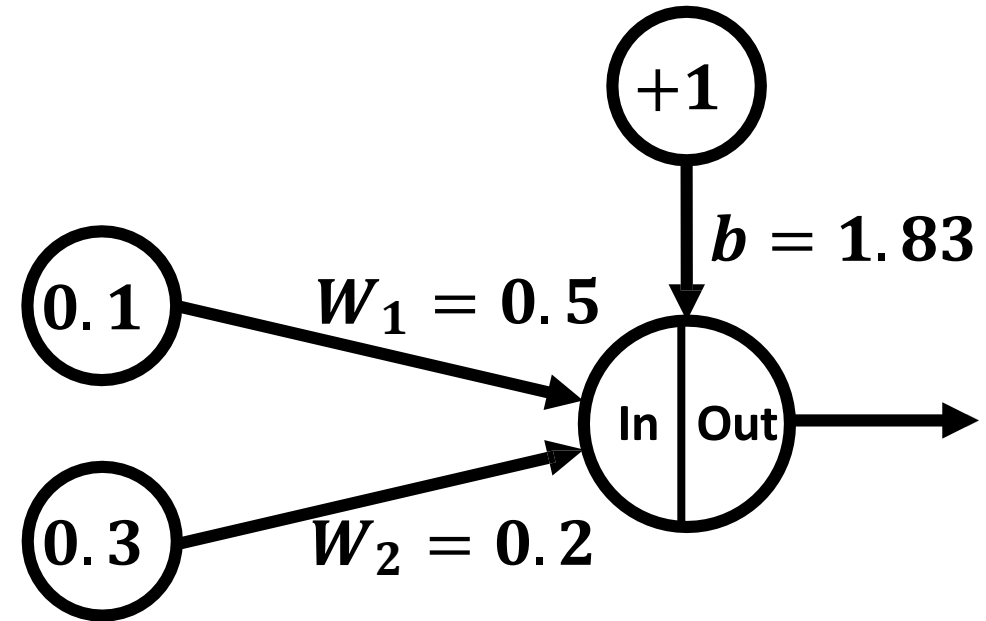
Network Training: Prediction Error

- After getting the predicted outputs, next is to measure the **prediction error** of the network.
- We can use the squared error function defined as follows:

$$E = \frac{1}{2} (\textit{desired} - \textit{predicted})^2$$

- Based on the predicted output, the prediction error is:

$$E = \frac{1}{2} (0.03 - 0.874)^2 = \frac{1}{2} (-0.844)^2 = \frac{1}{2} (0.713) = \mathbf{0.357}$$



How to Minimize Prediction Error?

- There is a prediction error and it should be minimized until reaching an acceptable error.

What should we do in order to minimize the error?

- There must be something to change in order to minimize the error. In our example, the only parameter to change is the weight.

How to update the weights?

- We can use the weights update equation:

$$W_{new} = W_{old} + \eta[d - Y]X$$

Weights Update Equation

- We can use the weights update equation:

$$\mathbf{W}_{new} = \mathbf{W}_{old} + \eta[d - Y]\mathbf{X}$$

- $\mathbf{W}_{new}$: new updated weights.
- $\mathbf{W}_{old}$: current weights. [1.83, 0.5, 0.2]
- η : network learning rate. 0.01
- d : desired output. 0.03
- Y : predicted output. 0.874
- $\mathbf{X}$: current input at which the network made false prediction. [+1, 0.1, 0.3]

Weights Update Equation

$$W_{new} = W_{old} + \eta[d - Y]X$$

$$= [1.83, 0.5, 0.2] + 0.01[0.03 - 0.874][+1, 0.1, 0.3]$$

$$= [1.83, 0.5, 0.2] + -0.0084[+1, 0.1, 0.3]$$

$$= [1.83, 0.5, 0.2] + [-0.0084, -0.00084, -0.0025]$$

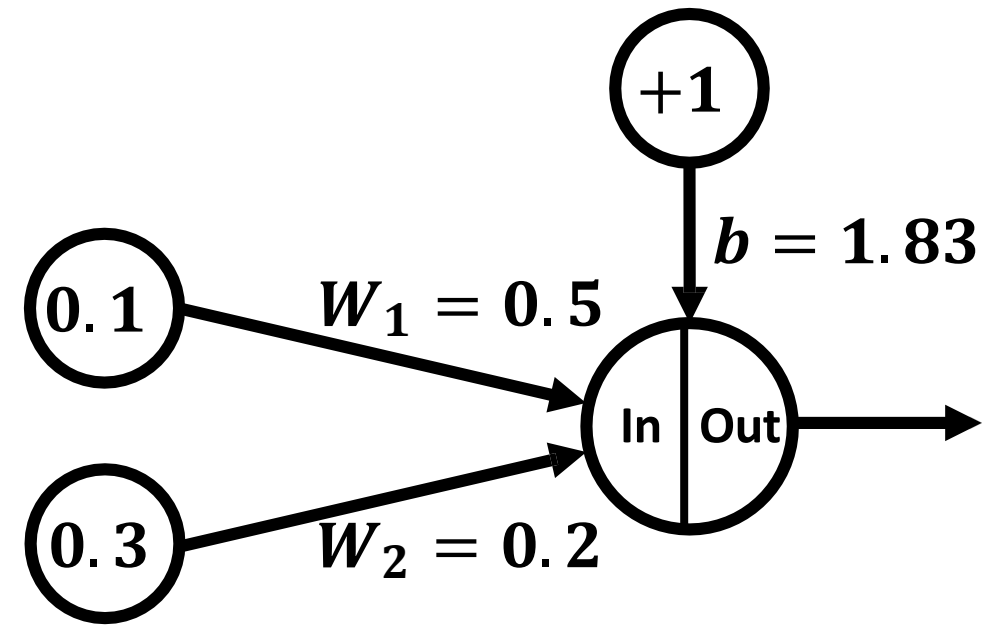
$$= [1.822, 0.499, 0.198]$$

Weights Update Equation

- The new weights are:

W_{1new}	W_{2new}	b_{new}
0.198	0.499	1.822

- Based on the new weights, the network will be re-trained.

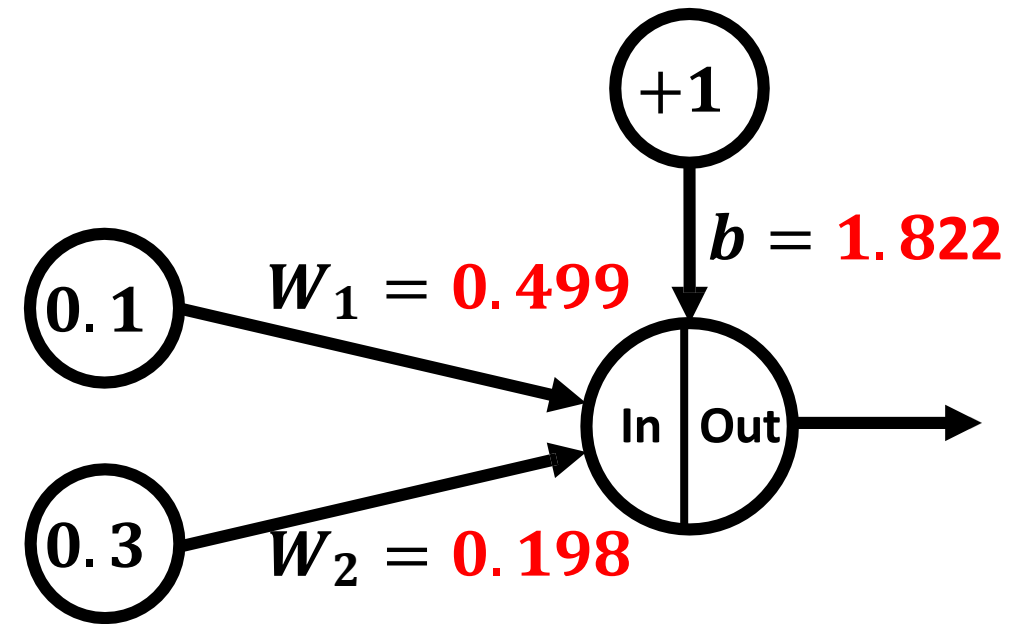


Weights Update Equation

- The new weights are:

W_{1new}	W_{2new}	b_{new}
0.198	0.499	1.822

- Based on the new weights, the network will be re-trained.
- Continue these operations until prediction error reaches an acceptable value.
 - Updating weights.
 - Retraining network.
 - Calculating prediction error.



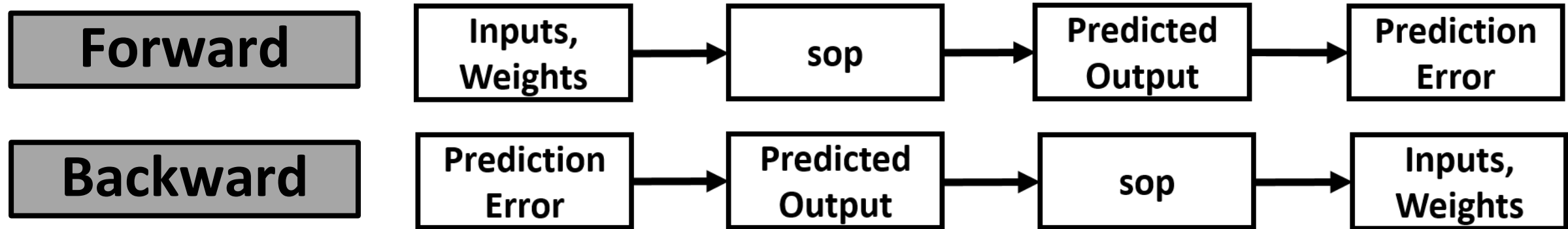
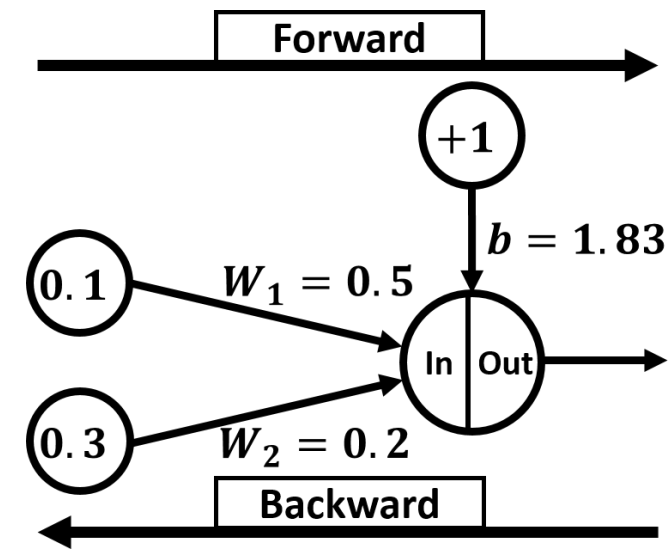
Why Backpropagation Algorithm is Important?



- The **backpropagation algorithm** is used to answer these questions and understand **effect** of each weight over the prediction error.

Forward Vs. Backward Passes

- When training a neural network, there are two passes: forward and backward.



- The goal of the backward pass is to know how each weight affects the total error. In other words, how changing the weights changes the prediction error?

Backward Pass

- Let us work with a simpler example:

$$Y = X^2Z + H$$

- How to answer this question: **What is the effect on the output Y given a change in variable X?**
- This question is answered using derivatives. Derivative of Y wrt X ($\frac{\partial Y}{\partial X}$) will tell us the effect of changing the variable X over the output Y.

Calculating Derivatives

$$Y = X^2Z + H$$

- The derivative $\frac{\partial Y}{\partial X}$ can be calculated as follows:

$$\frac{\partial Y}{\partial X} = \frac{\partial}{\partial X} (X^2Z + H)$$

- Based on these two derivative rules:

Square

$$\frac{\partial}{\partial X} (X^2) = 2X$$

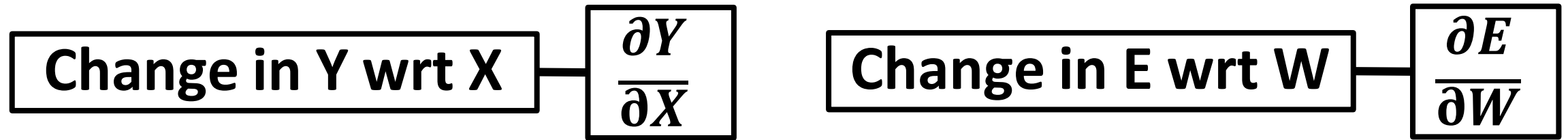
Constant

$$\frac{\partial}{\partial X} (C) = 0$$

- The result will be:

$$\frac{\partial Y}{\partial X} = 2XZ + 0 = 2XZ$$

Prediction Error – Weight Derivative



$$E = \frac{1}{2} (\textit{desired} - \textit{predicted})^2$$

Prediction Error – Weight Derivative

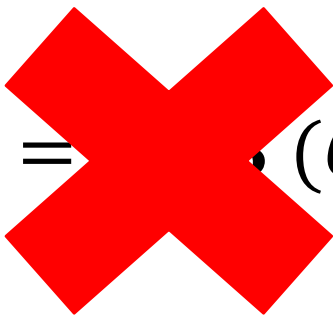
$$E = \frac{1}{2} (\textit{desired} - \textit{predicted})^2$$

Prediction Error – Weight Derivative

$$E = \frac{1}{2} (\textit{desired} - \textit{predicted})^2$$

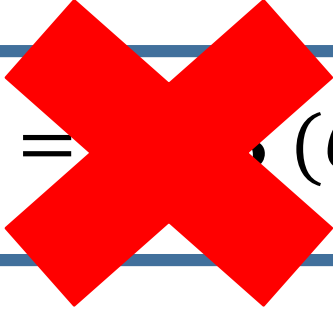
Prediction Error – Weight Derivative

$$E = \frac{1}{2} (\textit{desired} - \textit{predicted})^2$$

desired =  *(Constant)*

Prediction Error – Weight Derivative

$$E = \frac{1}{2} (\textit{desired} - \textit{predicted})^2$$


$$\textit{desired} = \text{Constant} \quad \textit{Predicted} = f(s) = \frac{1}{1 + e^{-s}}$$

Prediction Error – Weight Derivative

$$E = \frac{1}{2} (\textit{desired} - \textit{predicted})^2$$

$$\textit{desired} = 0.03 \text{ (Constant)} \quad \textit{Predicted} = f(s) = \frac{1}{1 + e^{-s}}$$

$$E = \frac{1}{2} \left(\textit{desired} - \frac{1}{1 + e^{-s}} \right)^2$$

Prediction Error – Weight Derivative

$$E = \frac{1}{2} (\textit{desired} - \textit{predicted})^2$$

$$\textit{desired} = 0.03 \text{ (Constant)} \quad \textit{Predicted} = f(s) = \frac{1}{1 + e^{-s}}$$

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Prediction Error – Weight Derivative

$$E = \frac{1}{2} (\textit{desired} - \textit{predicted})^2$$

$$\textit{desired} = 0.03 \text{ (Constant)} \quad \textit{Predicted} = f(s) = \frac{1}{1 + e^{-s}}$$

$$E = \frac{1}{2} \left(\textit{desired} - \frac{1}{1 + e^{-s}} \right)^2$$

$$\textcolor{red}{s} = X_1 * W_1 + X_2 * W_2 + b$$

Prediction Error – Weight Derivative

$$E = \frac{1}{2} (\textit{desired} - \textit{predicted})^2$$

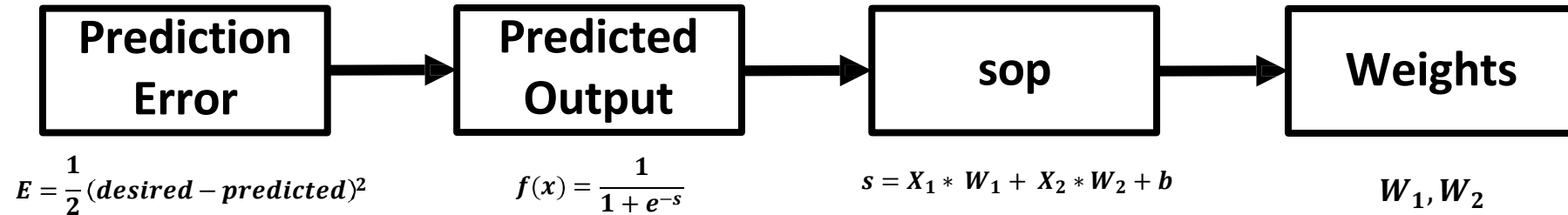
$$\textit{desired} = 0.03 \text{ (Constant)} \quad \textit{Predicted} = f(s) = \frac{1}{1 + e^{-s}}$$

$$E = \frac{1}{2} \left(\textit{desired} - \frac{1}{1 + e^{-s}} \right)^2$$

$$s = X_1 * W_1 + X_2 * W_2 + b$$

$$E = \frac{1}{2} \left(\textit{desired} - \frac{1}{1 + e^{-(X_1 * W_1 + X_2 * W_2 + b)}} \right)^2$$

Multivariate Chain Rule

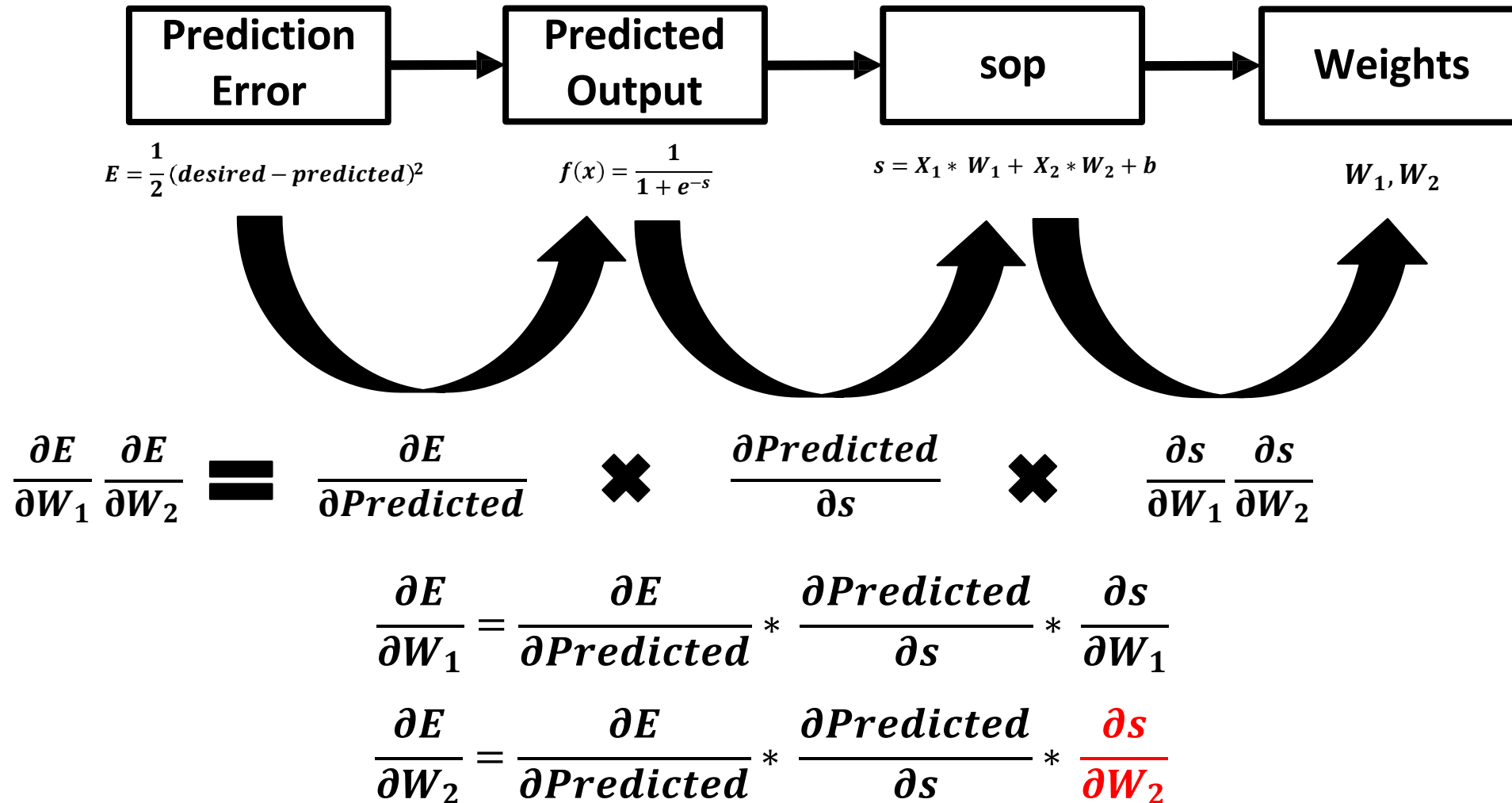


Chain Rule

$$E = \frac{1}{2} \left(\text{desired} - \frac{1}{1 + e^{-(X_1 * W_1 + X_2 * W_2 + b)}} \right)^2$$

$$\frac{\partial E}{\partial W} = \frac{\partial}{\partial W} \left(\frac{1}{2} \left(\text{desired} - \frac{1}{1 + e^{-(X_1 * W_1 + X_2 * W_2 + b)}} \right)^2 \right)$$

Multivariate Chain Rule



Let's calculate these individual partial derivatives.

Error-Predicted ($\frac{\partial E}{\partial Predicted}$) Partial Derivative

$$E = \frac{1}{2} (desired - predicted)^2$$

$$\begin{aligned}\frac{\partial E}{\partial Predicted} &= \frac{\partial}{\partial Predicted} \left(\frac{1}{2} (desired - predicted)^2 \right) \\ &= 2 * \frac{1}{2} (desired - predicted)^{2-1} * (0 - 1) \\ &= (desired - predicted) * (-1) \\ &= predicted - desired\end{aligned}$$

Substitution

$$\frac{\partial E}{\partial Predicted} = predicted - desired = 0.874 - 0.03$$

$$\frac{\partial E}{\partial Predicted} = 0.844$$

Predicted-sop ($\frac{\partial Predicted}{\partial s}$) Partial Derivative

$$\text{Predicted} = \frac{1}{1 + e^{-s}}$$

$$\frac{\partial Predicted}{\partial s} = \frac{\partial}{\partial s} \left(\frac{1}{1 + e^{-s}} \right)$$

$$\frac{\partial Predicted}{\partial s} = \frac{1}{1 + e^{-s}} \left(1 - \frac{1}{1 + e^{-s}} \right)$$

Substitution

$$\frac{\partial Predicted}{\partial s} = \frac{1}{1 + e^{-s}} \left(1 - \frac{1}{1 + e^{-s}} \right) = \frac{1}{1 + e^{-1.94}} \left(1 - \frac{1}{1 + e^{-1.94}} \right)$$

$$= \frac{1}{1 + 0.144} \left(1 - \frac{1}{1 + 0.144} \right)$$

$$= \frac{1}{1.144} \left(1 - \frac{1}{1.144} \right)$$

$$= 0.874(1 - 0.874)$$

$$= 0.874(0.126)$$

$$\frac{\partial Predicted}{\partial s} = 0.11$$

Sop- W_1 ($\frac{\partial s}{\partial W_1}$) Partial Derivative

$$s = X_1 * W_1 + X_2 * W_2 + b$$

$$\frac{\partial s}{\partial W_1} = \frac{\partial}{\partial W_1} (X_1 * W_1 + X_2 * W_2 + b)$$

$$= 1 * X_1 * (W_1)^{(1-1)} + 0 + 0$$

$$= X_1 * (W_1)^{(0)}$$

$$= X_1(1)$$

$$\frac{\partial s}{\partial W_1} = X_1$$

Substitution

$$\frac{\partial s}{\partial W_1} = X_1$$

$$\frac{\partial s}{\partial W_1} = 0.1$$

Sop- W_1 ($\frac{\partial s}{\partial W_2}$) Partial Derivative

$$s = X_1 * W_1 + X_2 * W_2 + b$$

$$\frac{\partial s}{\partial W_2} = \frac{\partial}{\partial W_2} (X_1 * W_1 + X_2 * W_2 + b)$$

$$= 0 + 1 * X_2 * (W_2)^{(1-1)} + 0$$

$$= X_2 * (W_2)^{(0)}$$

$$= X_2(1)$$

$$\frac{\partial s}{\partial W_2} = X_2$$

Substitution

$$\frac{\partial s}{\partial W_2} = X_2 = 0.3$$

$$\frac{\partial s}{\partial W_2} = 0.3$$

Error- W_1 ($\frac{\partial E}{\partial W_1}$) Partial Derivative

- After calculating each individual derivative, we can multiply all of them to get the desired relationship between the prediction error and each weight.

Calculated Derivatives

$$\frac{\partial E}{\partial Predicted} = 0.844$$
$$\frac{\partial Predicted}{\partial s} = 0.11$$
$$\frac{\partial s}{\partial W_1} = 0.1$$

$$\frac{\partial E}{\partial W_1} = \frac{\partial E}{\partial Predicted} * \frac{\partial Predicted}{\partial s} * \frac{\partial s}{\partial W_1}$$

$$\frac{\partial E}{\partial W_1} = 0.844 * 0.11 * 0.1$$

$$\frac{\partial E}{\partial W_1} = 0.01$$

Error- W_2 ($\frac{\partial E}{\partial W_2}$) Partial Derivative

Calculated Derivatives

$$\frac{\partial E}{\partial \text{Predicted}} = 0.844$$

$$\frac{\partial \text{Predicted}}{\partial s} = 0.11$$

$$\frac{\partial s}{\partial W_2} = 0.3$$

$$\frac{\partial E}{\partial W_2} = \frac{\partial E}{\partial \text{Predicted}} * \frac{\partial \text{Predicted}}{\partial s} * \frac{\partial s}{\partial W_2}$$

$$\frac{\partial E}{\partial W_2} = 0.844 * 0.11 * 0.3$$

$$\frac{\partial E}{\partial W_2} = 0.03$$

Interpreting Derivatives

$$\frac{\partial E}{\partial W_1} = 0.01$$

$$\frac{\partial E}{\partial W_2} = 0.03$$

- There are two useful pieces of information from the derivatives calculated previously.

Derivative Sign

Positive

Increasing/decreasing weight
increases/decreases error.

Negative

Increasing/decreasing weight
decreases/increases error.

Derivative Magnitude

Positive
Sign

Increasing/decreasing weight by P
increases/decreases error by MAG*P.

Negative
Sign

Increasing/decreasing weight by P
decreases/increases error by MAG*P.

In our example, because both $\frac{\partial E}{\partial W_1}$ and $\frac{\partial E}{\partial W_2}$ are positive, then we would like to decrease the weights in order to decrease the prediction error.

Updating Weights

- Each weight will be updated based on its derivative according to this equation:

$$W_{inew} = W_{iold} - \eta * \frac{\partial E}{\partial W_i}$$

Updating W_1

$$\begin{aligned} W_{1new} &= W_1 - \eta * \frac{\partial E}{\partial W_1} \\ &= 0.5 - 0.01 * 0.01 \end{aligned}$$

$$W_{1new} = 0.49991$$

Updating W_2

$$\begin{aligned} W_{2new} &= W_2 - \eta * \frac{\partial E}{\partial W_2} \\ &= 0.2 - 0.01 * 0.028 \end{aligned}$$

$$W_{2new} = 0.1997$$

Continue updating weights according to derivatives and re-train the network until reaching an acceptable error.

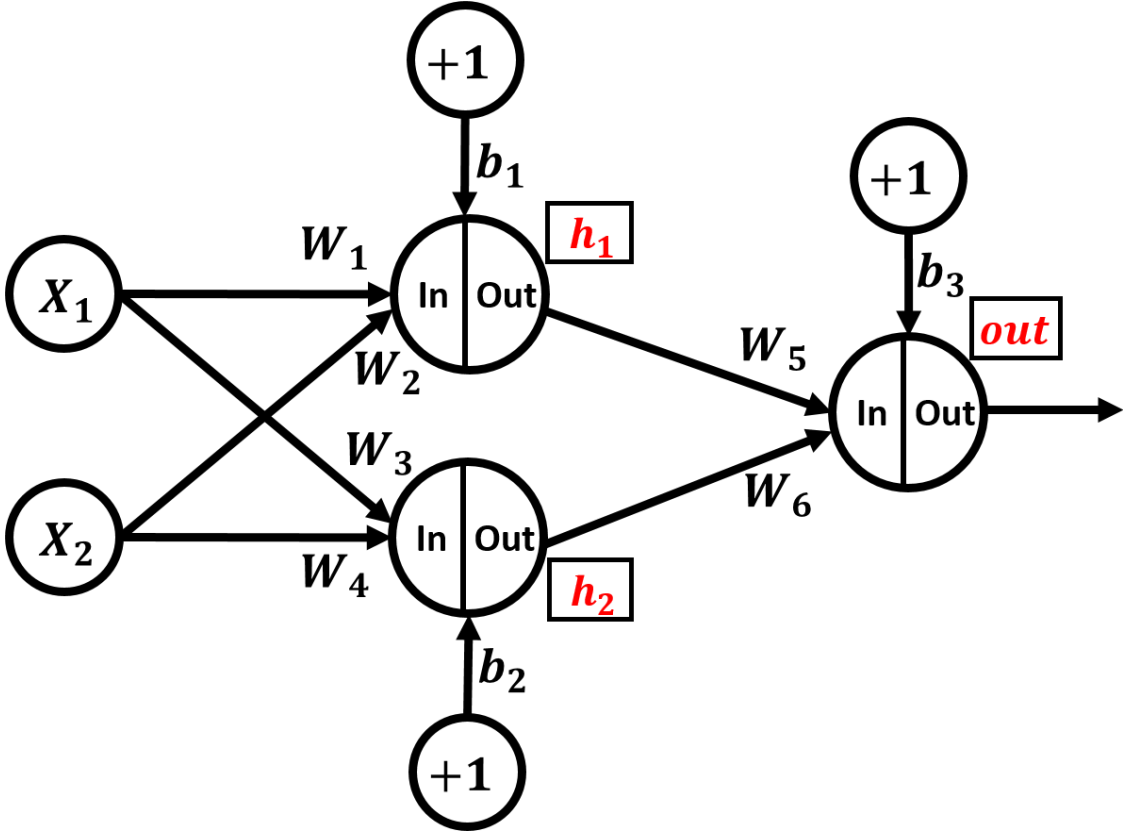
Second Example

Backpropagation for NN with Hidden Layer

ANN with Hidden Layer

Training Data

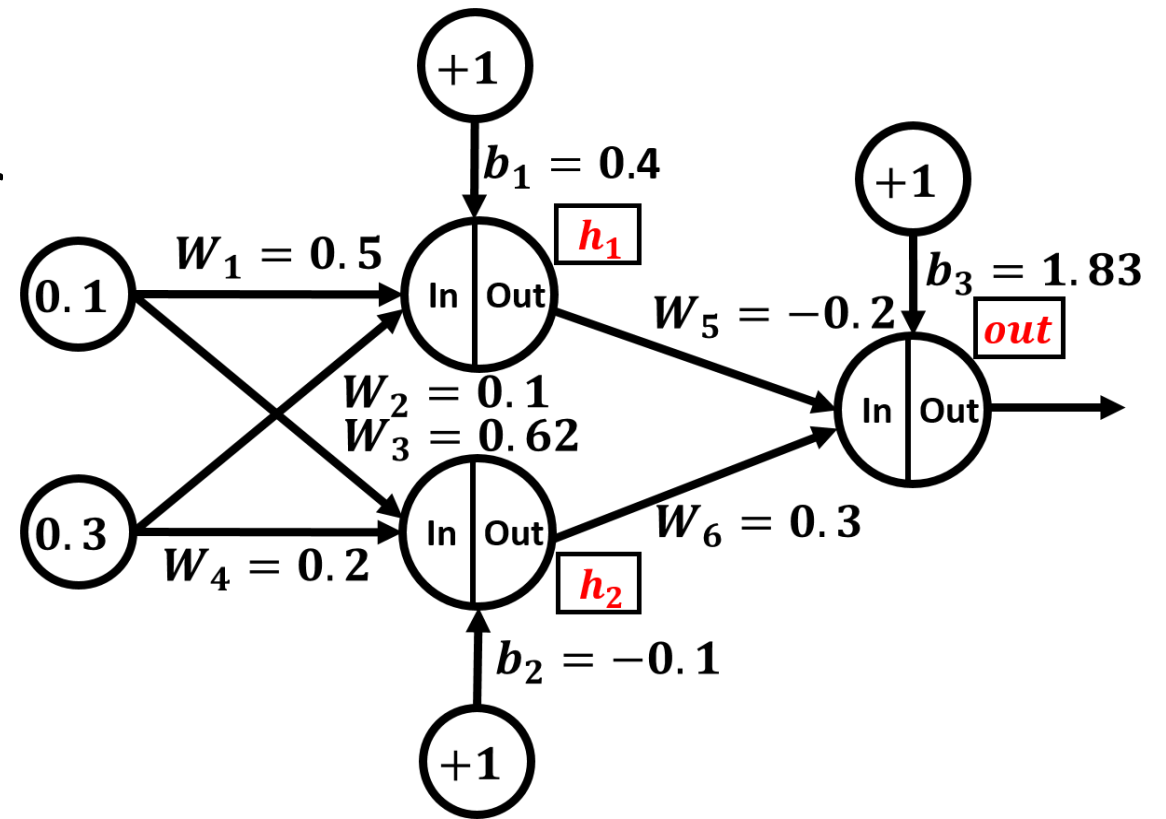
X_1	X_2	Output
0.1	0.3	0.03



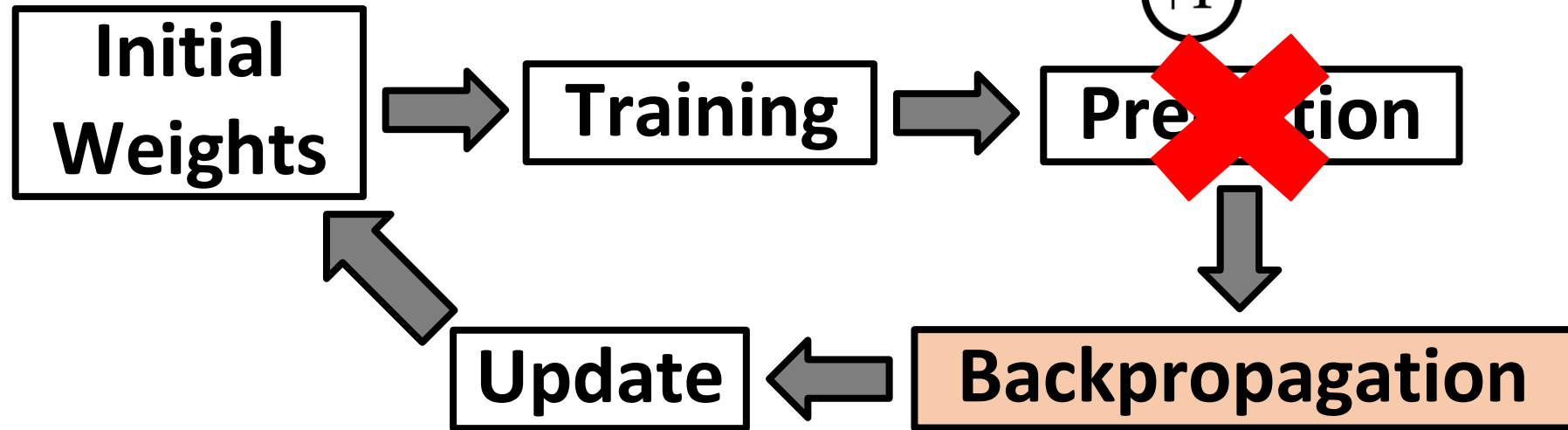
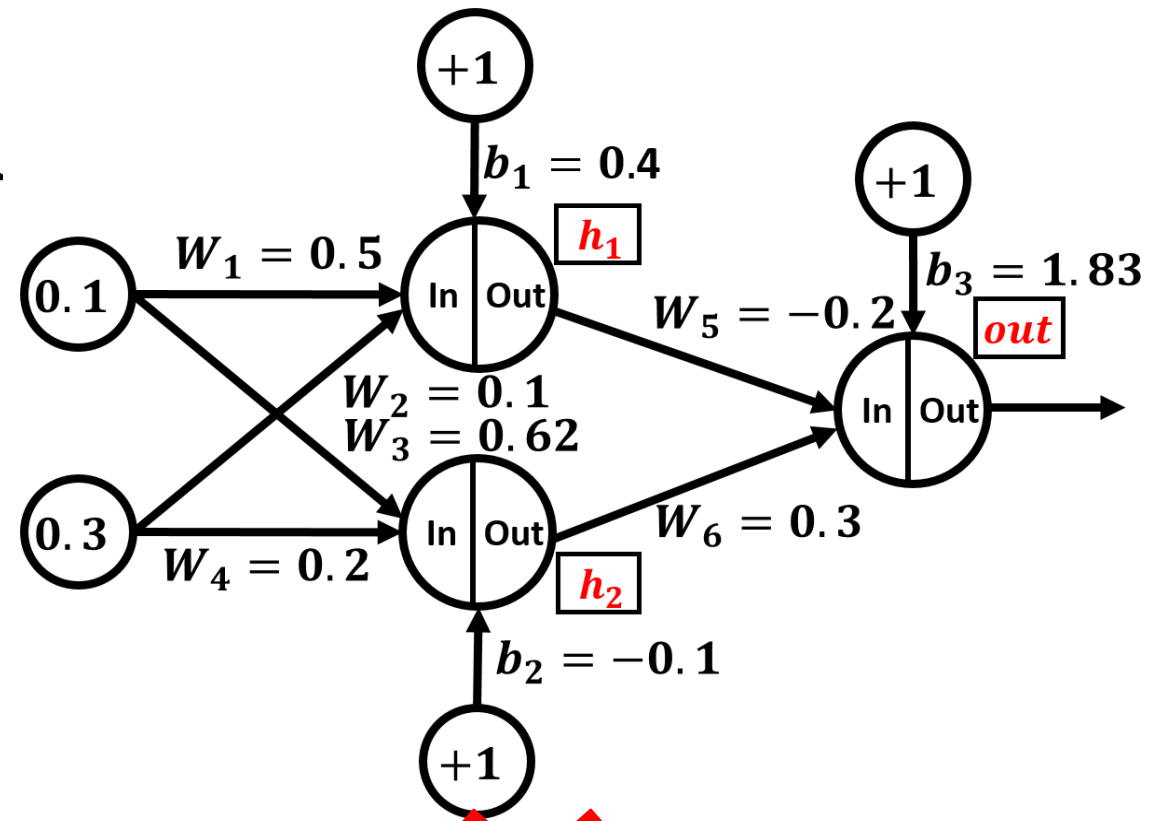
Initial Weights

W_1	W_2	W_3	W_4	W_5	W_6	b_1	b_2	b_3
0.5	0.1	0.62	0.2	-0.2	0.3	0.4	-0.1	1.83

ANN with Hidden Layer



ANN with Hidden Layer



Forward Pass – Hidden Layer Neurons

h_1

In

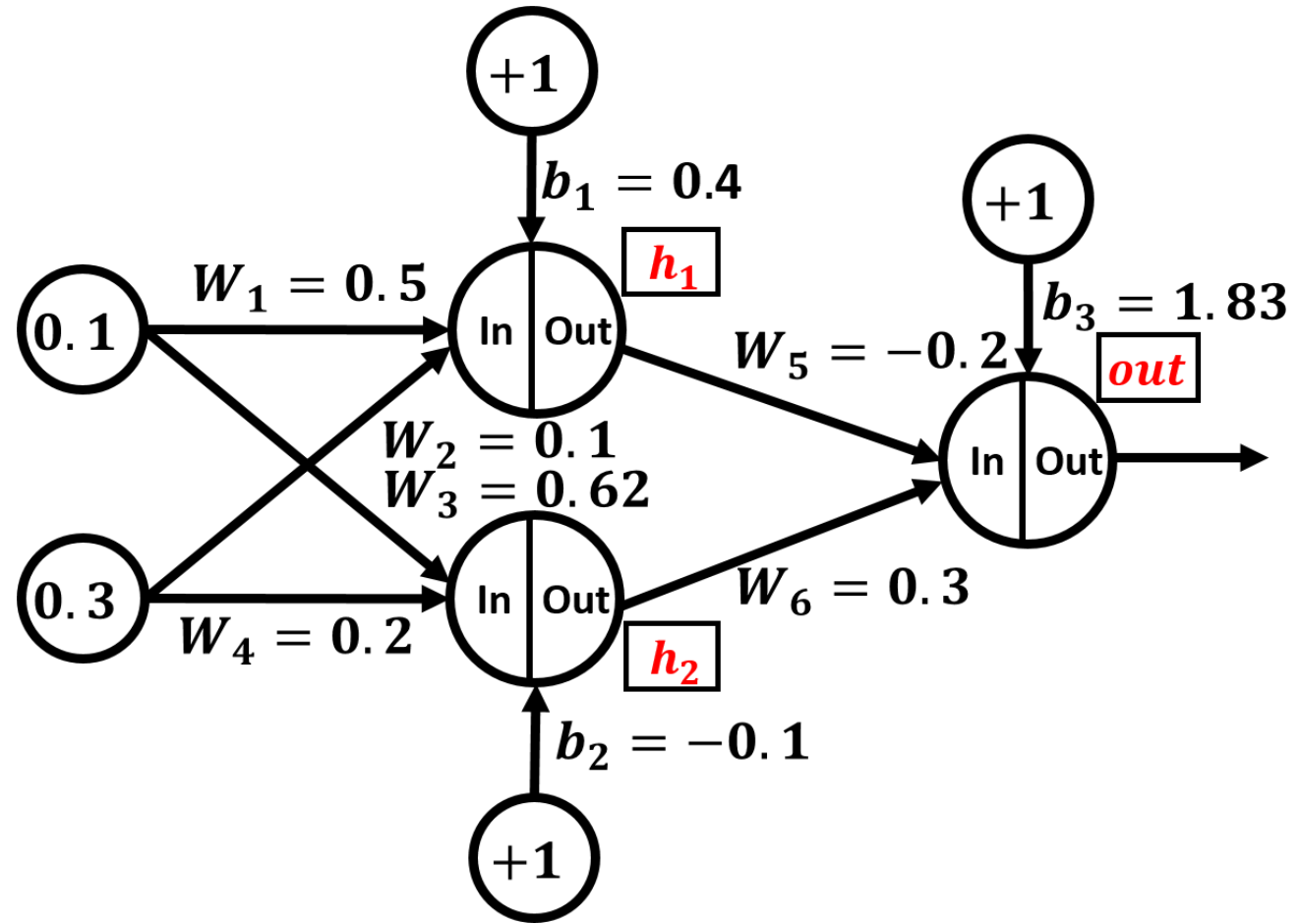
$$h_{1in} = X_1 * W_1 + X_2 * W_2 + b_1$$
$$= 0.1 * 0.5 + 0.3 * 0.1 + 0.4$$

$h_{1in} = 0.48$

Out

$$h_{1out} = \frac{1}{1 + e^{-h_{1in}}}$$
$$= \frac{1}{1 + e^{-0.48}}$$

$h_{1out} = 0.618$



Forward Pass – Hidden Layer Neurons

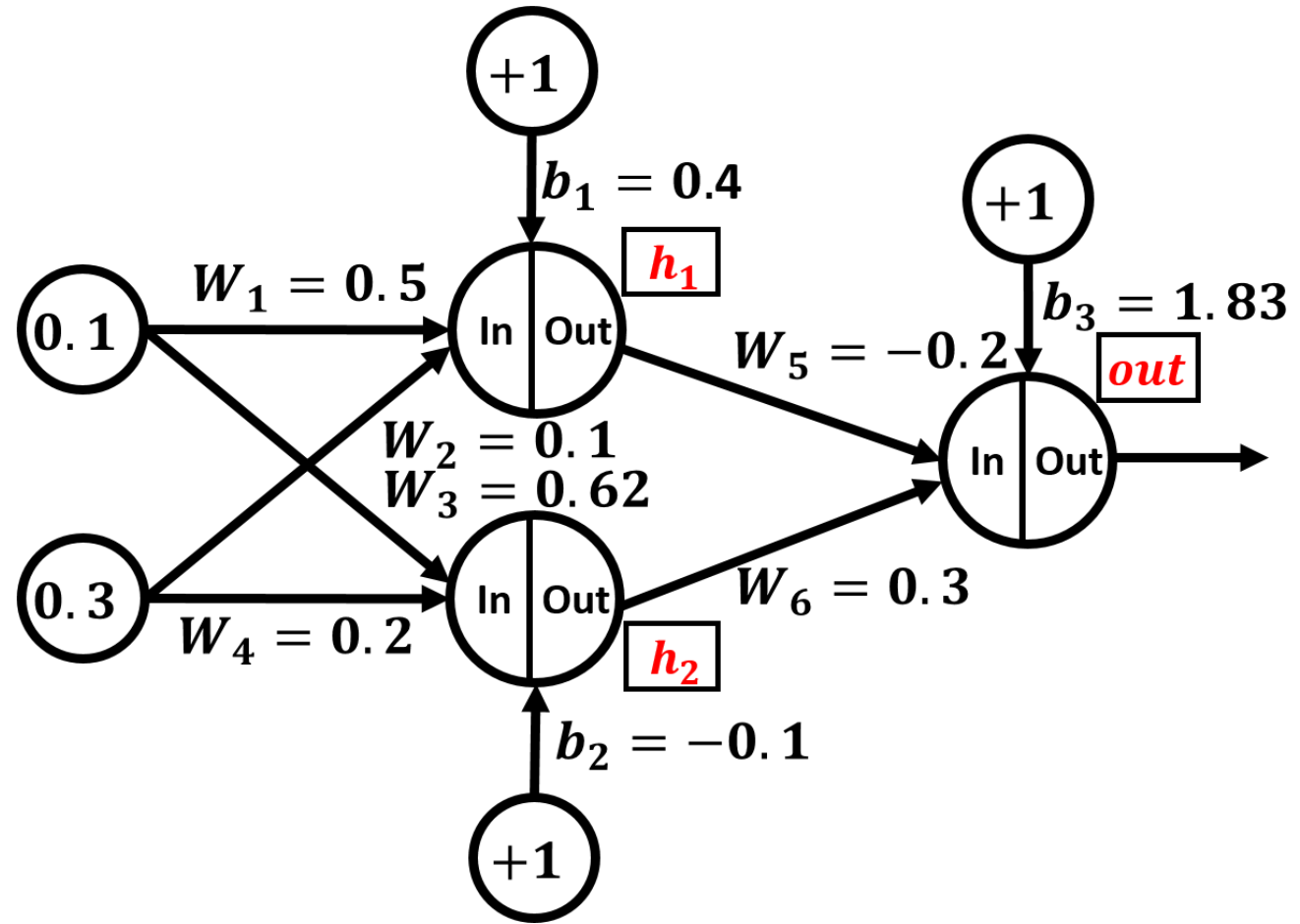
h_2

In

$$h_{2in} = X_1 * W_3 + X_2 * W_4 + b_2$$
$$= 0.1 * 0.62 + 0.3 * 0.2 - 0.1$$
$$h_{2in} = 0.022$$

Out

$$h_{2out} = \frac{1}{1 + e^{-h_{2in}}}$$
$$= \frac{1}{1 + e^{-0.022}}$$
$$h_{2out} = 0.506$$



Forward Pass – Output Layer Neuron

out

In

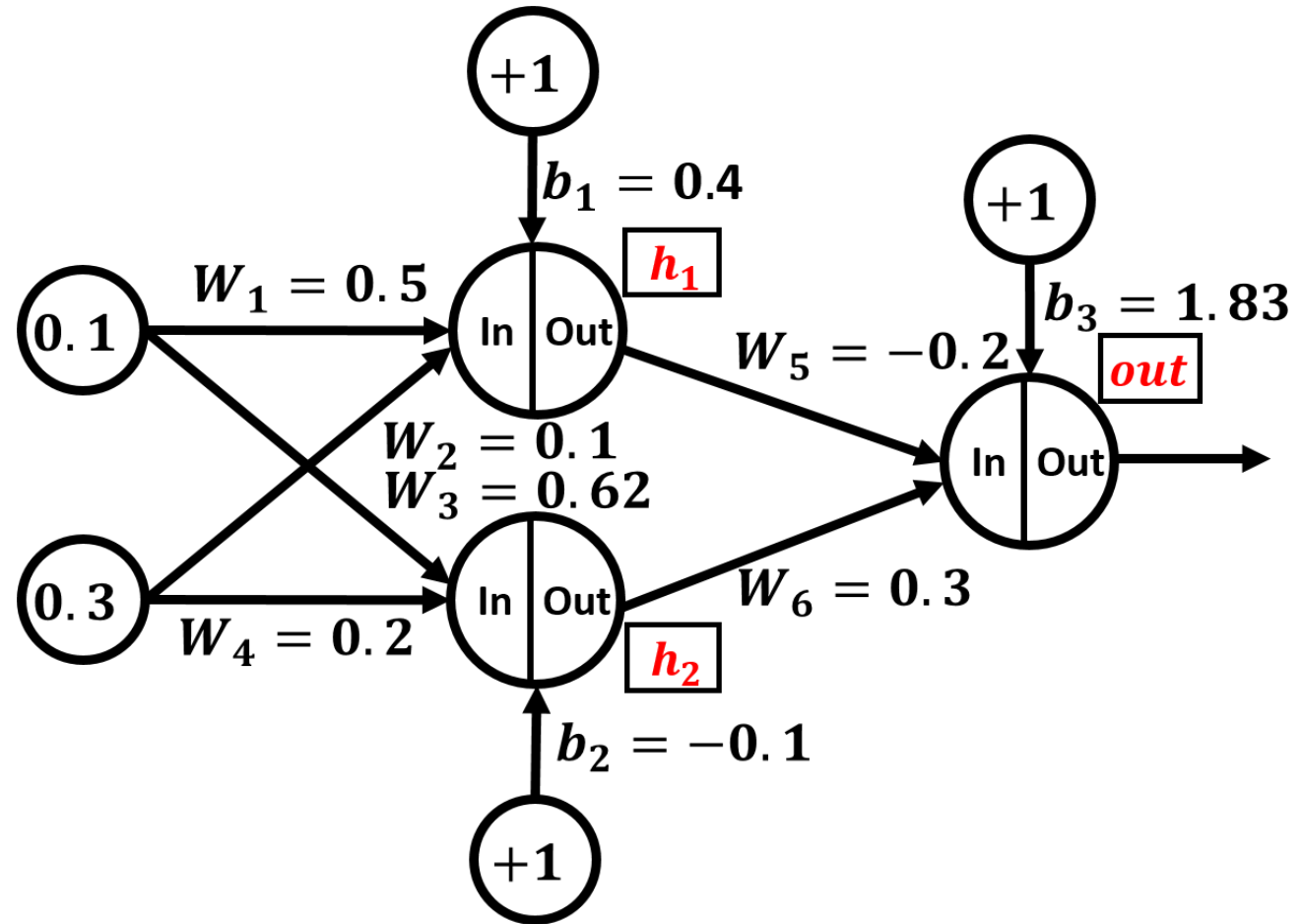
$$out_{in} = h_{1out} * W_5 + h_{2out} * W_6 + b_3$$
$$= 0.618 * -0.2 + 0.506 * 0.3 + 1.83$$

$$out_{in} = 1.858$$

Out

$$out_{out} = \frac{1}{1 + e^{-out_{in}}}$$
$$= \frac{1}{1 + e^{-1.858}}$$

$$out_{out} = 0.865$$



Forward Pass – Prediction Error

desired = 0.03

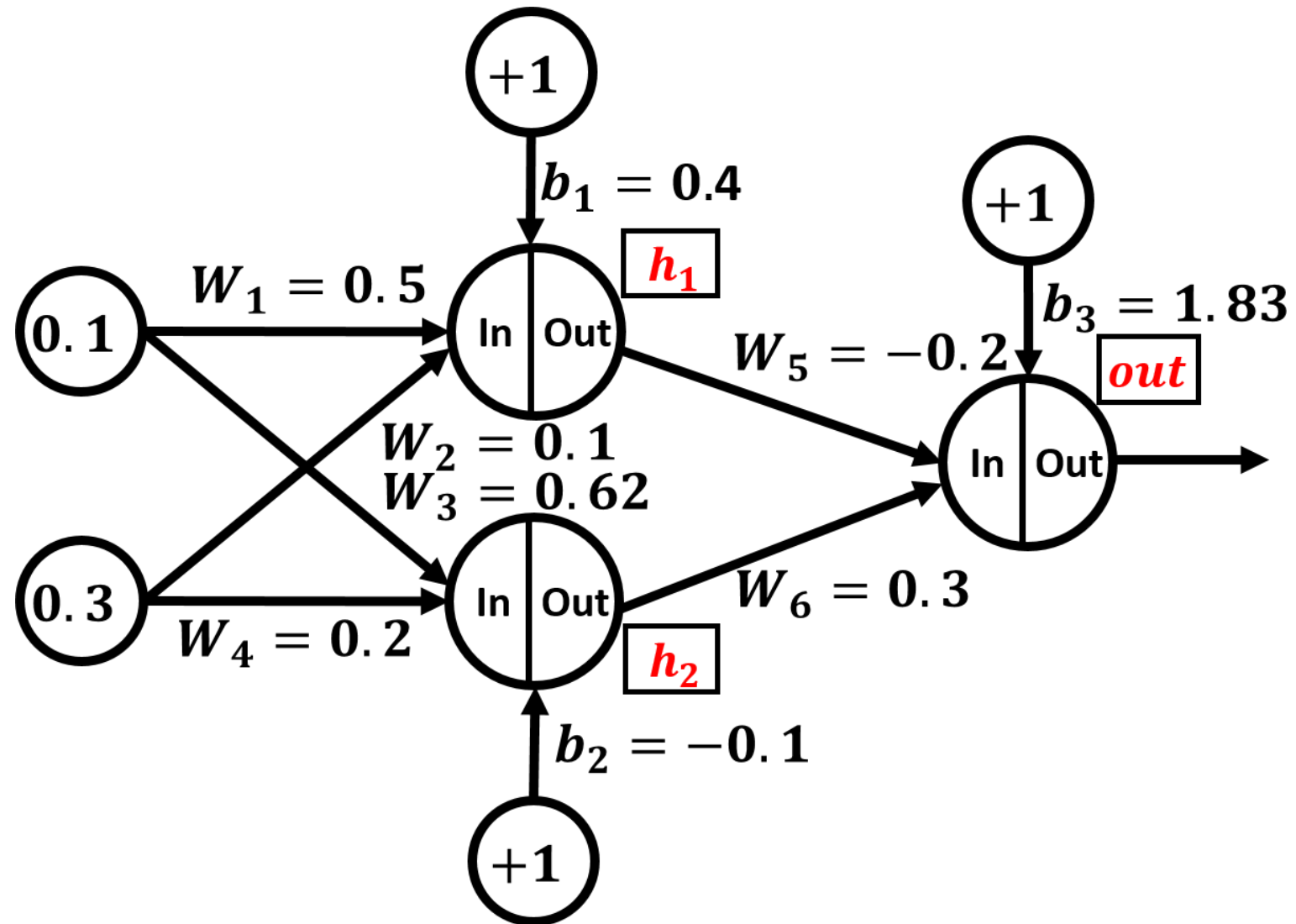
Predicted = out_{out} = 0.865

$$E = \frac{1}{2} (desired - out_{out})^2$$
$$= \frac{1}{2} (0.03 - 0.865)^2$$

$$E = 0.349$$

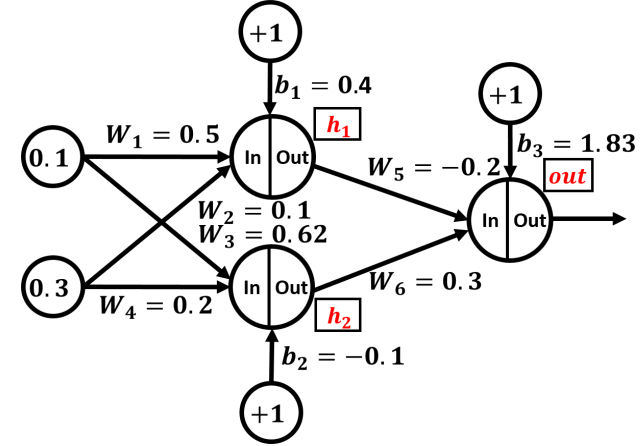
$$\frac{\partial E}{\partial W_1}, \frac{\partial E}{\partial W_2}, \frac{\partial E}{\partial W_3}, \frac{\partial E}{\partial W_4}, \frac{\partial E}{\partial W_5}, \frac{\partial E}{\partial W_6}$$

Partial Derivatives Calculation

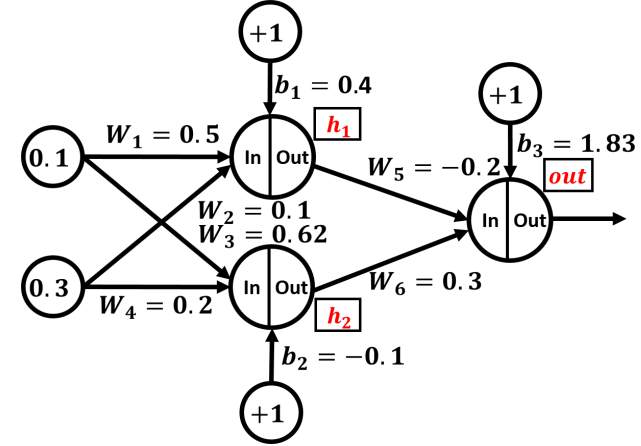


$E - W_5 \left(\frac{\partial E}{\partial W_5} \right)$ Partial Derivative

$$\frac{\partial E}{\partial W_5} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial W_5}$$



$E - W_5 \left(\frac{\partial E}{\partial W_5} \right)$ Partial Derivative



$$\frac{\partial E}{\partial W_5} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial W_5}$$

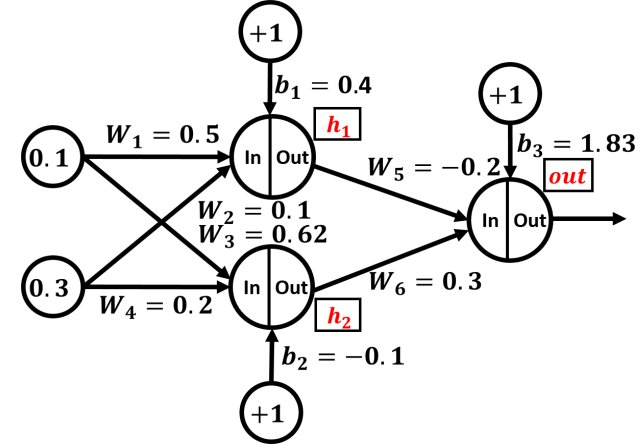
Partial Derivative

$$\begin{aligned} \frac{\partial E}{\partial out_{out}} &= \frac{\partial}{\partial out_{out}} \left(\frac{1}{2} (desired - out_{out})^2 \right) \\ &= 2 * \frac{1}{2} (desired - out_{out})^{2-1} * (0 - 1) \\ &= desired - out_{out} * (-1) \\ \frac{\partial E}{\partial out_{out}} &= out_{out} - desired \end{aligned}$$

Substitution

$$\begin{aligned} \frac{\partial E}{\partial out_{out}} &= out_{out} - desired = 0.865 - 0.03 \\ \frac{\partial E}{\partial out_{out}} &= 0.835 \end{aligned}$$

$E - W_5 \left(\frac{\partial E}{\partial W_5} \right)$ Partial Derivative



$$\frac{\partial E}{\partial W_5} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial W_5}$$

Partial Derivative

$$\frac{\partial out_{out}}{\partial out_{in}} = \frac{\partial}{\partial out_{in}} \left(\frac{1}{1 + e^{-out_{in}}} \right)$$

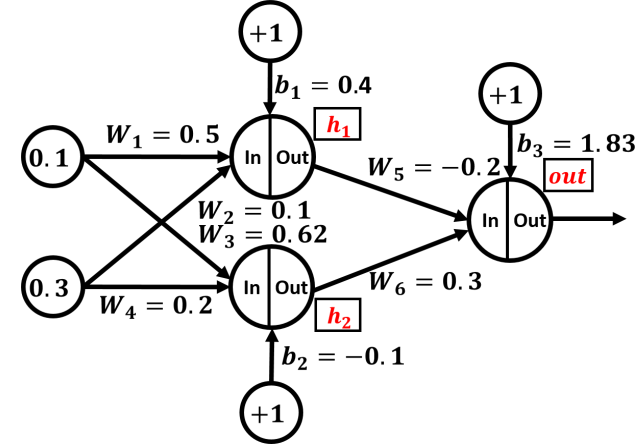
$$\frac{\partial out_{out}}{\partial out_{in}} = \left(\frac{1}{1 + e^{-out_{in}}} \right) \left(1 - \frac{1}{1 + e^{-out_{in}}} \right)$$

Substitution

$$\begin{aligned} \frac{\partial out_{out}}{\partial out_{in}} &= \left(\frac{1}{1 + e^{-1.858}} \right) \left(1 - \frac{1}{1 + e^{-1.858}} \right) \\ &= \left(\frac{1}{1.56} \right) \left(1 - \frac{1}{1.56} \right) \\ &= (0.641)(1 - 0.641) = (0.641)(0.359) \end{aligned}$$

$$\frac{\partial out_{out}}{\partial out_{in}} = 0.23$$

$E - W_5 \left(\frac{\partial E}{\partial W_5} \right)$ Partial Derivative



$$\frac{\partial E}{\partial W_5} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial W_5}$$

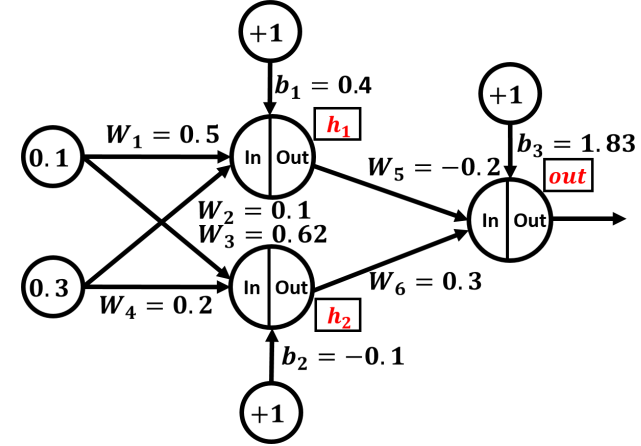
Partial Derivative

$$\begin{aligned} \frac{\partial out_{in}}{\partial W_5} &= \frac{\partial}{\partial W_5} (h_{1out} * W_5 + h_{2out} * W_6 + b_3) \\ &= 1 * h_{1out} * (W_5)^{1-1} + 0 + 0 \\ \frac{\partial out_{in}}{\partial W_5} &= h_{1out} \end{aligned}$$

Substitution

$$\begin{aligned} \frac{\partial out_{in}}{\partial W_5} &= h_{1out} \\ \frac{\partial out_{in}}{\partial W_5} &= 0.618 \end{aligned}$$

$E - W_5 \left(\frac{\partial E}{\partial W_5} \right)$ Partial Derivative



$$\frac{\partial E}{\partial W_5} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial W_5}$$

$$\frac{\partial E}{\partial out_{out}} = 0.835$$

$$\frac{\partial out_{out}}{\partial out_{in}} = 0.23$$

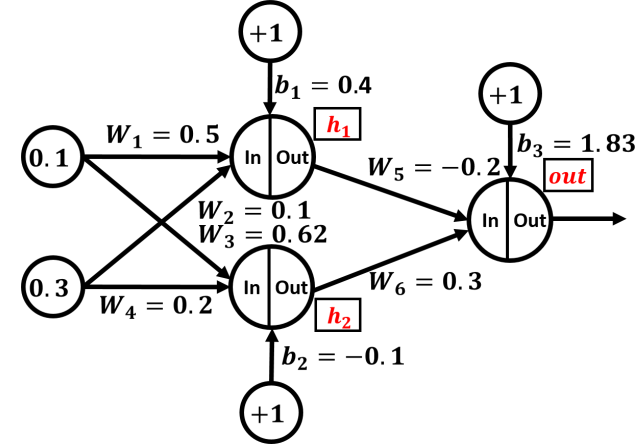
$$\frac{\partial out_{in}}{\partial W_5} = 0.618$$

$$\frac{\partial E}{\partial W_5} = 0.835 * 0.23 * 0.618$$

$$\frac{\partial E}{\partial W_5} = 0.119$$

$E - W_6 \left(\frac{\partial E}{\partial W_6} \right)$ Partial Derivative

$$\frac{\partial E}{\partial W_6} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial W_6}$$

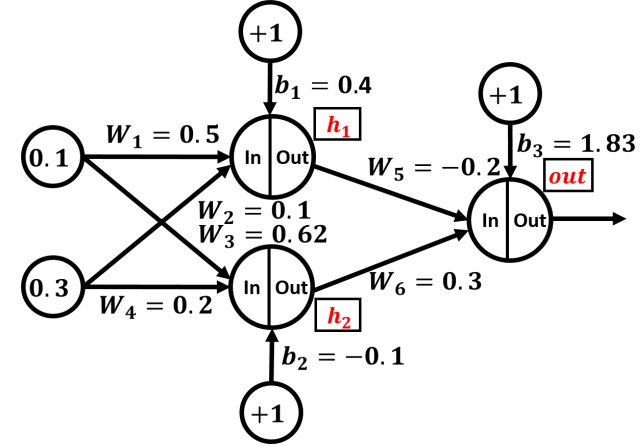


$E - W_6 \left(\frac{\partial E}{\partial W_6} \right)$ Partial Derivative

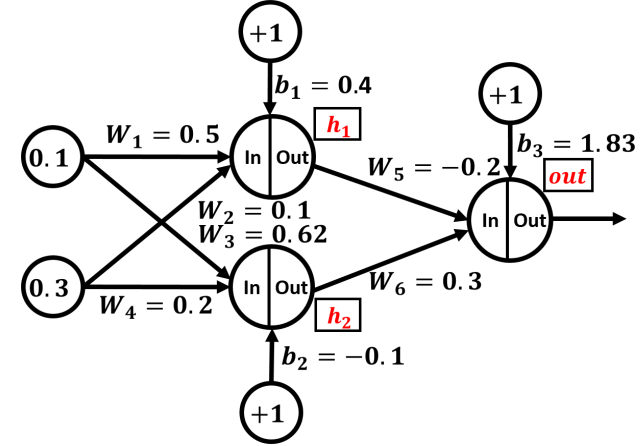
$$\frac{\partial E}{\partial W_6} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial W_6}$$

$$\frac{\partial E}{\partial out_{out}} = 0.835$$

$$\frac{\partial out_{out}}{\partial out_{in}} = 0.23$$



$E - W_6 \left(\frac{\partial E}{\partial W_6} \right)$ Partial Derivative



$$\frac{\partial E}{\partial W_5} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial W_6}$$

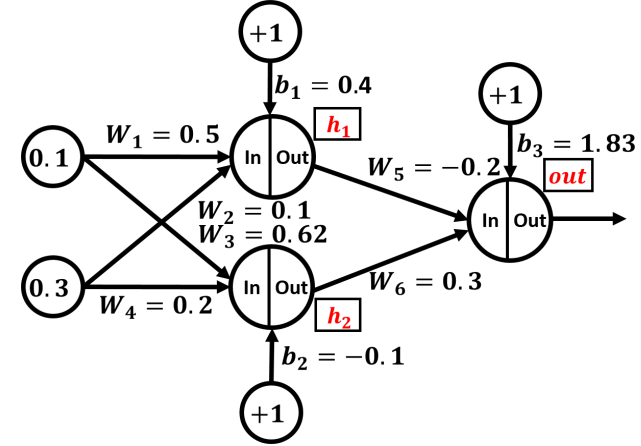
Partial Derivative

$$\begin{aligned} \frac{\partial out_{in}}{\partial W_6} &= \frac{\partial}{\partial W_6} (h_{1out} * W_5 + h_{2out} * W_6 + b_3) \\ &= 0 + 1 * h_{2out} * (W_6)^{1-1} + 0 \\ \frac{\partial out_{in}}{\partial W_6} &= h_{2out} \end{aligned}$$

Substitution

$$\begin{aligned} \frac{\partial out_{in}}{\partial W_6} &= h_{2out} \\ \frac{\partial out_{in}}{\partial W_6} &= 0.506 \end{aligned}$$

$E - W_6 \left(\frac{\partial E}{\partial W_6} \right)$ Partial Derivative



$$\frac{\partial E}{\partial W_6} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial W_6}$$

$$\frac{\partial E}{\partial out_{out}} = 0.835$$

$$\frac{\partial out_{out}}{\partial out_{in}} = 0.23$$

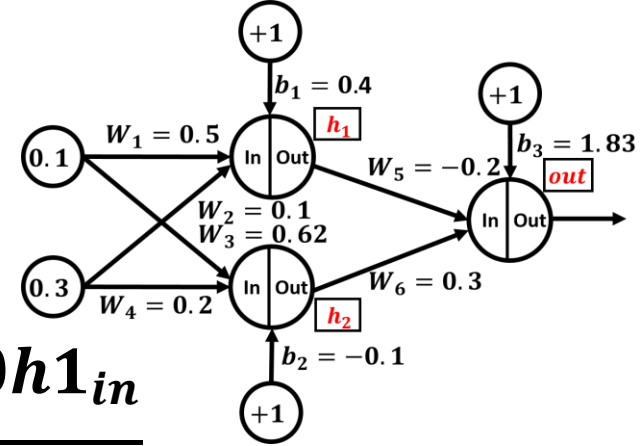
$$\frac{\partial out_{in}}{\partial W_6} = 0.506$$

$$\frac{\partial E}{\partial W_6} = 0.835 * 0.23 * 0.506$$

$$\frac{\partial E}{\partial W_6} = 0.097$$

$E - W_1 \left(\frac{\partial E}{\partial W_1} \right)$ Partial Derivative

$$\frac{\partial E}{\partial W_1} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h1_{out}} * \frac{\partial h1_{out}}{\partial h1_{in}} * \frac{\partial h1_{in}}{\partial W_1}$$

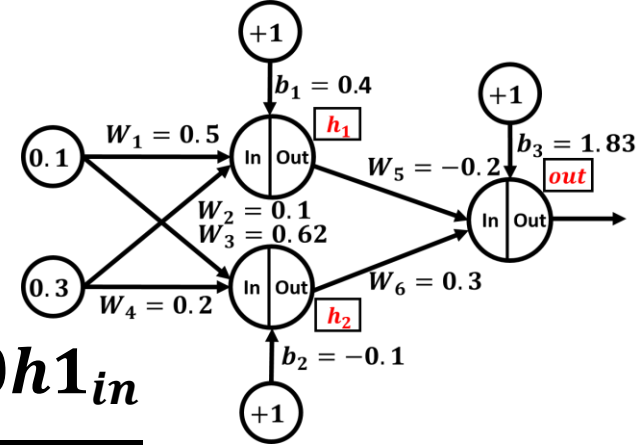


$E - W_1 \left(\frac{\partial E}{\partial W_1} \right)$ Partial Derivative

$$\frac{\partial E}{\partial W_1} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h1_{out}} * \frac{\partial h1_{out}}{\partial h1_{in}} * \frac{\partial h1_{in}}{\partial W_1}$$

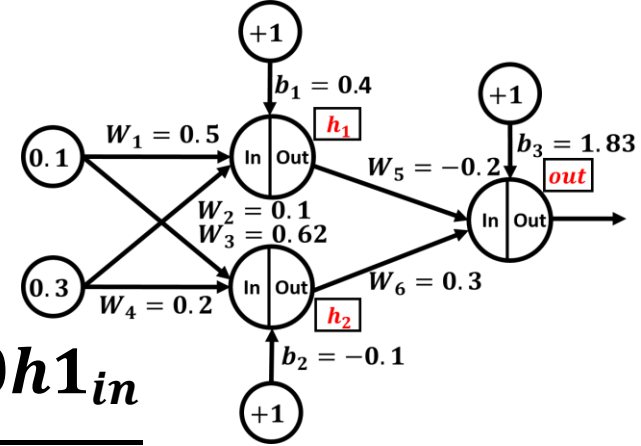
$$\frac{\partial E}{\partial out_{out}} = 0.835$$

$$\frac{\partial out_{out}}{\partial out_{in}} = 0.23$$



$E - W_1 \left(\frac{\partial E}{\partial W_1} \right)$ Partial Derivative

$$\frac{\partial E}{\partial W_1} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h1_{out}} * \frac{\partial h1_{out}}{\partial h1_{in}} * \frac{\partial h1_{in}}{\partial W_1}$$



Partial Derivative

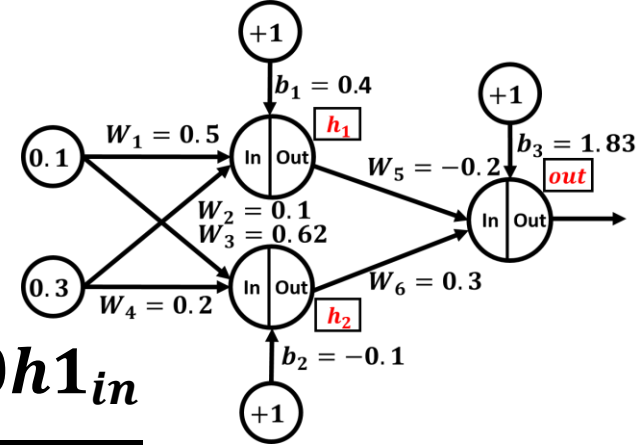
$$\begin{aligned} \frac{\partial out_{in}}{\partial h1_{out}} &= \frac{\partial}{\partial h1_{out}} (h1_{out} * W_5 + h2_{out} * W_6 + b_3) \\ &= (h1_{out})^{1-1} * W_5 + 0 + 0 \\ \frac{\partial out_{in}}{\partial h1_{out}} &= W_5 \end{aligned}$$

Substitution

$$\begin{aligned} \frac{\partial out_{in}}{\partial h1_{out}} &= W_5 \\ \frac{\partial out_{in}}{\partial h1_{out}} &= -0.2 \end{aligned}$$

$E - W_1 \left(\frac{\partial E}{\partial W_1} \right)$ Partial Derivative

$$\frac{\partial E}{\partial W_1} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h1_{out}} * \frac{\partial h1_{out}}{\partial h1_{in}} * \frac{\partial h1_{in}}{\partial W_1}$$



Partial Derivative

$$\frac{\partial h1_{out}}{\partial h1_{in}} = \frac{\partial}{\partial h1_{in}} \left(\frac{1}{1 + e^{-h1_{in}}} \right)$$

$$\frac{\partial h1_{out}}{\partial h1_{in}} = \left(\frac{1}{1 + e^{-h1_{in}}} \right) \left(1 - \frac{1}{1 + e^{-h1_{in}}} \right)$$

Substitution

$$\frac{\partial h1_{out}}{\partial h1_{in}} = \left(\frac{1}{1 + e^{-h1_{in}}} \right) \left(1 - \frac{1}{1 + e^{-h1_{in}}} \right)$$

$$= \left(\frac{1}{1 + e^{-0.48}} \right) \left(1 - \frac{1}{1 + e^{-0.48}} \right)$$

$$\frac{\partial h2_{out}}{\partial h2_{in}} = 0.236$$

$E - W_1 \left(\frac{\partial E}{\partial W_1} \right)$ Partial Derivative

$$\frac{\partial E}{\partial W_1} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h1_{out}} * \frac{\partial h1_{out}}{\partial h1_{in}} * \frac{\partial h1_{in}}{\partial W_1}$$

Partial Derivative

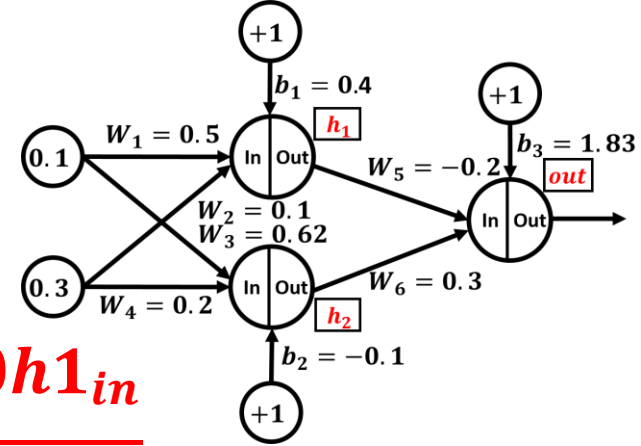
$$\begin{aligned} \frac{\partial h1_{in}}{\partial W_1} &= \frac{\partial}{\partial W_1} (X_1 * W_1 + X_2 * W_2 + b_1) \\ &= X_1 * (W_1)^{1-1} + 0 + 0 \end{aligned}$$

$$\frac{\partial h1_{in}}{\partial W_1} = X_1$$

Substitution

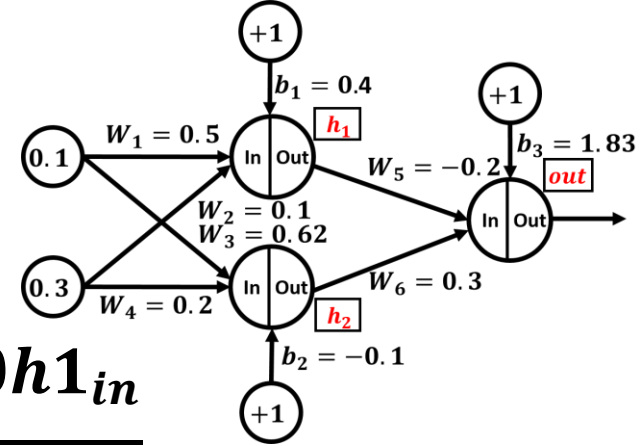
$$\frac{\partial h1_{in}}{\partial W_1} = X_1$$

$$\frac{\partial h1_{in}}{\partial W_1} = 0.1$$



$$E - W_1 \left(\frac{\partial E}{\partial W_1} \right) \text{Partial Derivative}$$

$$\frac{\partial E}{\partial W_1} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h1_{out}} * \frac{\partial h1_{out}}{\partial h1_{in}} * \frac{\partial h1_{in}}{\partial W_1}$$



$$\frac{\partial E}{\partial out_{out}} = 0.835$$

$$\frac{\partial out_{out}}{\partial out_{in}} = 0.23$$

$$\frac{\partial out_{in}}{\partial h1_{out}} = -0.2$$

$$\frac{\partial h2_{out}}{\partial h2_{in}} = 0.236$$

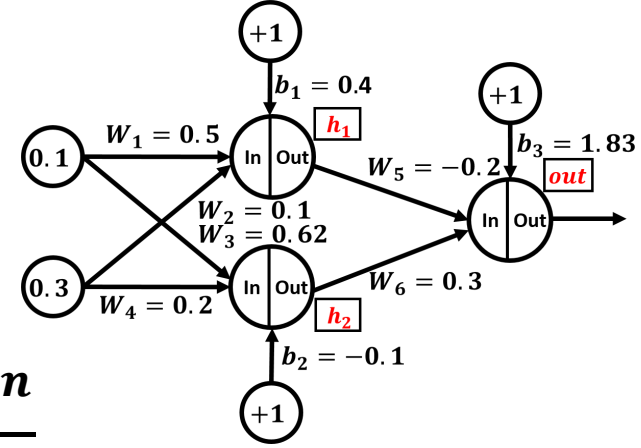
$$\frac{\partial h1_{in}}{\partial W_1} = 0.1$$

$$\frac{\partial E}{\partial W_1} = 0.835 * 0.23 * -0.2 * 0.236 * 0.1$$

$$\frac{\partial E}{\partial W_1} = -0.001$$

$E - W_2 \left(\frac{\partial E}{\partial W_2} \right)$ Partial Derivative:

$$\frac{\partial E}{\partial W_2} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h1_{out}} * \frac{\partial h1_{out}}{\partial h1_{in}} * \frac{\partial h1_{in}}{\partial W_2}$$



$E - W_2 \left(\frac{\partial E}{\partial W_2} \right)$ Partial Derivative:

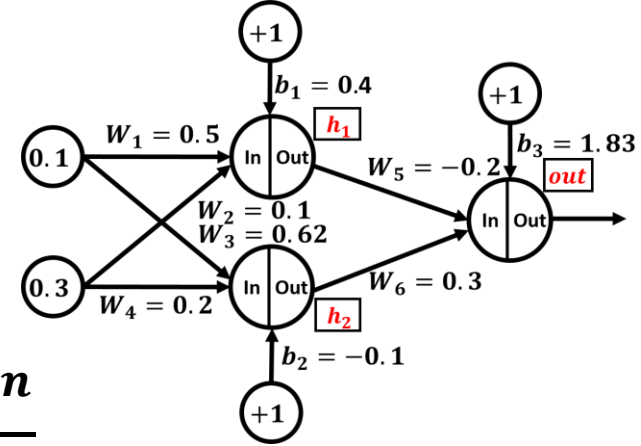
$$\frac{\partial E}{\partial W_2} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h1_{out}} * \frac{\partial h1_{out}}{\partial h1_{in}} * \frac{\partial h1_{in}}{\partial W_2}$$

$$\frac{\partial E}{\partial out_{out}} = 0.835$$

$$\frac{\partial out_{out}}{\partial out_{in}} = 0.23$$

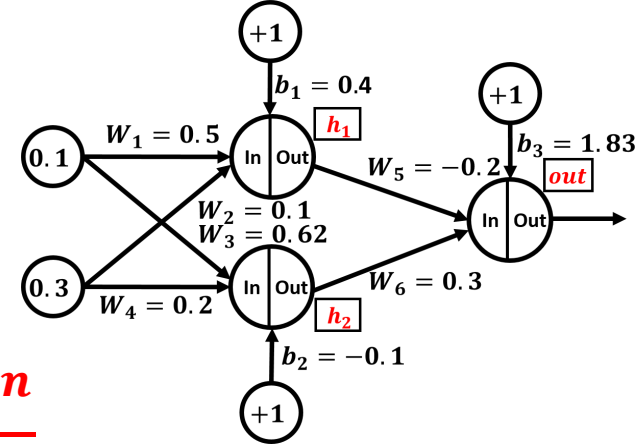
$$\frac{\partial out_{in}}{\partial h1_{out}} = -0.2$$

$$\frac{\partial h2_{out}}{\partial h2_{in}} = 0.236$$



$E - W_2 \left(\frac{\partial E}{\partial W_2} \right)$ Partial Derivative:

$$\frac{\partial E}{\partial W_2} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h1_{out}} * \frac{\partial h1_{out}}{\partial h1_{in}} * \frac{\partial h1_{in}}{\partial W_2}$$



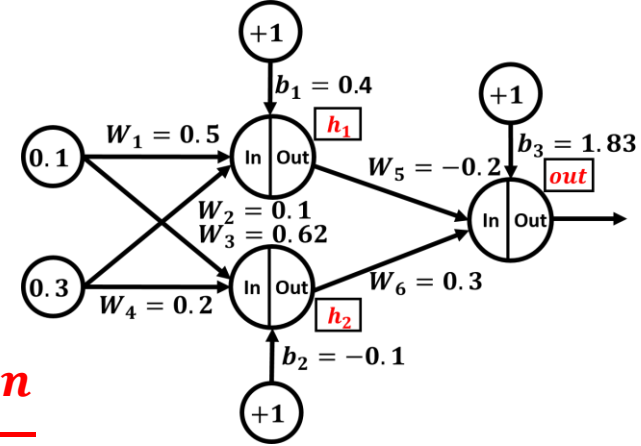
Partial Derivative

$$\begin{aligned} \frac{\partial h1_{in}}{\partial W_2} &= \frac{\partial}{\partial W_2} (X_1 * W_1 + X_2 * W_2 + b_1) \\ &= 0 + X_2 * (W_2)^{1-1} + 0 \\ \frac{\partial h1_{in}}{\partial W_2} &= X_2 \end{aligned}$$

Substitution

$$\begin{aligned} \frac{\partial h1_{in}}{\partial W_2} &= X_2 \\ \frac{\partial h1_{in}}{\partial W_2} &= 0.3 \end{aligned}$$

$E - W_2 \left(\frac{\partial E}{\partial W_2} \right)$ Partial Derivative:



$$\frac{\partial E}{\partial W_2} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h1_{out}} * \frac{\partial h1_{out}}{\partial h1_{in}} * \frac{\partial h1_{in}}{\partial W_2}$$

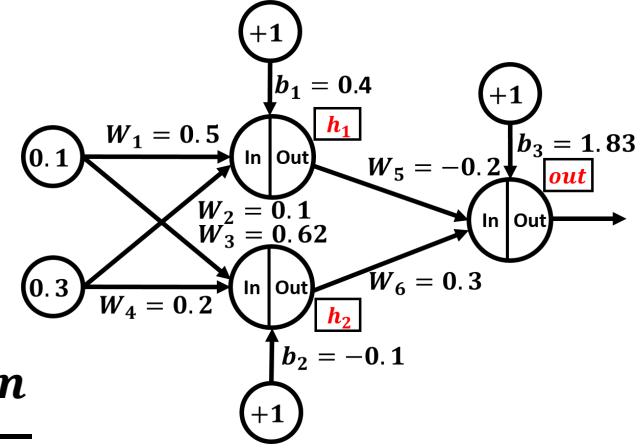
$$\frac{\partial E}{\partial out_{out}} = 0.835 \quad \frac{\partial out_{out}}{\partial out_{in}} = 0.23 \quad \frac{\partial out_{in}}{\partial h1_{out}} = -0.2 \quad \frac{\partial h2_{out}}{\partial h2_{in}} = 0.236 \quad \frac{\partial h1_{in}}{\partial W_2} = 0.3$$

$$\frac{\partial E}{\partial W_2} = 0.835 * 0.23 * -0.2 * 0.236 * 0.3$$

$$\frac{\partial E}{\partial W_2} = -.003$$

$E - W_3 \left(\frac{\partial E}{\partial W_3} \right)$ Partial Derivative:

$$\frac{\partial E}{\partial W_3} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h2_{out}} * \frac{\partial h2_{out}}{\partial h2_{in}} * \frac{\partial h2_{in}}{\partial W_3}$$

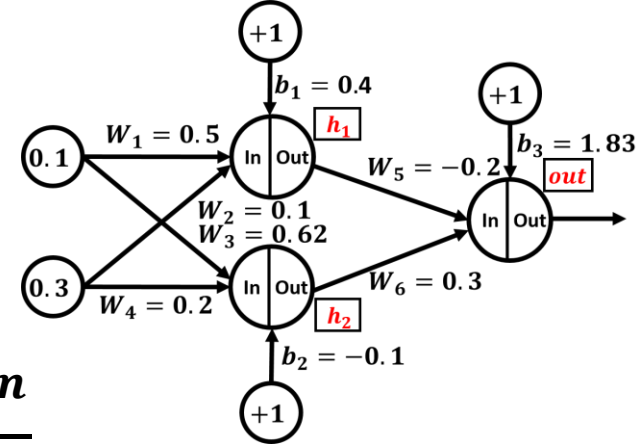


$E - W_3 \left(\frac{\partial E}{\partial W_3} \right)$ Partial Derivative:

$$\frac{\partial E}{\partial W_3} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h2_{out}} * \frac{\partial h2_{out}}{\partial h2_{in}} * \frac{\partial h2_{in}}{\partial W_3}$$

$$\frac{\partial E}{\partial out_{out}} = 0.835$$

$$\frac{\partial out_{out}}{\partial out_{in}} = 0.23$$



$E - W_3 \left(\frac{\partial E}{\partial W_3} \right)$ Partial Derivative:

$$\frac{\partial E}{\partial W_3} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h2_{out}} * \frac{\partial h2_{out}}{\partial h2_{in}} * \frac{\partial h2_{in}}{\partial W_3}$$

Partial Derivative

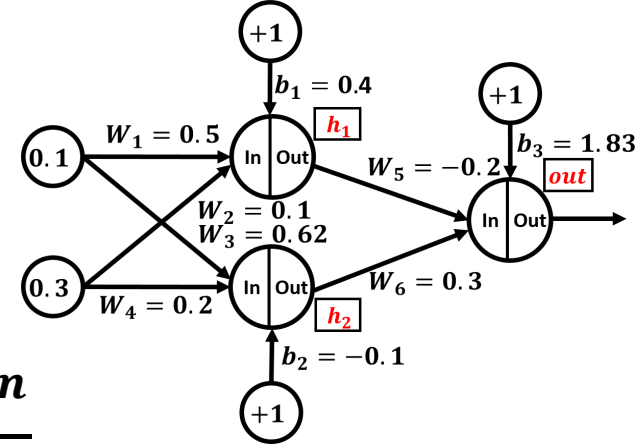
$$\begin{aligned} \frac{\partial out_{in}}{\partial h2_{out}} &= \frac{\partial}{\partial h2_{out}} (h1_{out} * W_5 + h2_{out} * W_6 + b_3) \\ &= 0 + (h2_{out})^{1-1} * W_6 + 0 \end{aligned}$$

$$\frac{\partial out_{in}}{\partial h2_{out}} = W_6$$

Substitution

$$\frac{\partial out_{in}}{\partial h2_{out}} = W_6$$

$$\frac{\partial out_{in}}{\partial h2_{out}} = 0.3$$



$E - W_3 \left(\frac{\partial E}{\partial W_3} \right)$ Partial Derivative:

$$\frac{\partial E}{\partial W_3} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h2_{out}} * \frac{\partial h2_{out}}{\partial h2_{in}} * \frac{\partial h2_{in}}{\partial W_3}$$

Partial Derivative

$$\frac{\partial h2_{out}}{\partial h2_{in}} = \frac{\partial}{\partial h2_{in}} \left(\frac{1}{1 + e^{-h2_{in}}} \right)$$

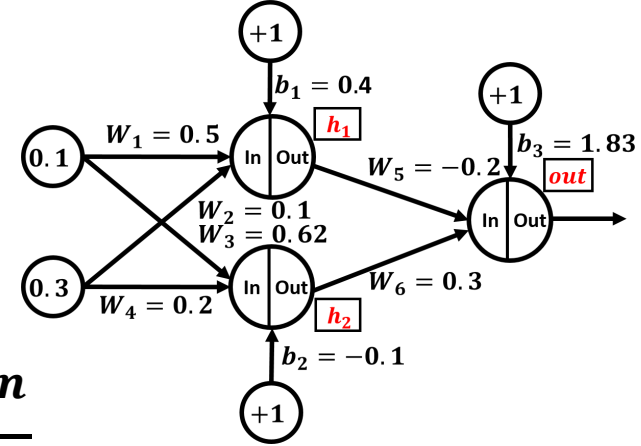
$$\frac{\partial h2_{out}}{\partial h2_{in}} = \left(\frac{1}{1 + e^{-h2_{in}}} \right) \left(1 - \frac{1}{1 + e^{-h2_{in}}} \right)$$

Substitution

$$\frac{\partial h2_{out}}{\partial h2_{in}} = \left(\frac{1}{1 + e^{-h2_{in}}} \right) \left(1 - \frac{1}{1 + e^{-h2_{in}}} \right)$$

$$= \left(\frac{1}{1 + e^{-0.022}} \right) \left(1 - \frac{1}{1 + e^{-0.022}} \right)$$

$$\frac{\partial h2_{out}}{\partial h2_{in}} = 0.25$$



$E - W_3 \left(\frac{\partial E}{\partial W_3} \right)$ Partial Derivative:

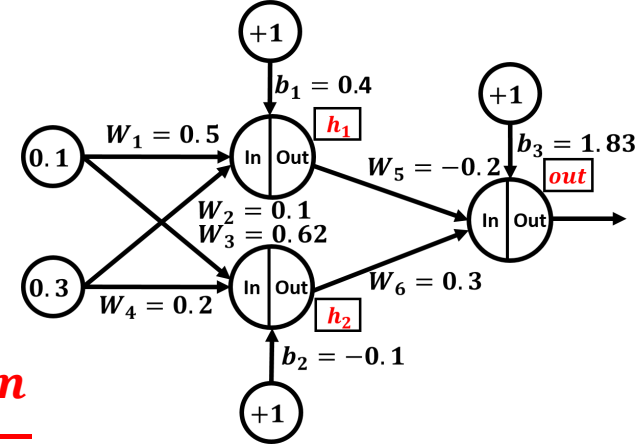
$$\frac{\partial E}{\partial W_3} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h2_{out}} * \frac{\partial h2_{out}}{\partial h2_{in}} * \frac{\partial h2_{in}}{\partial W_3}$$

Partial Derivative

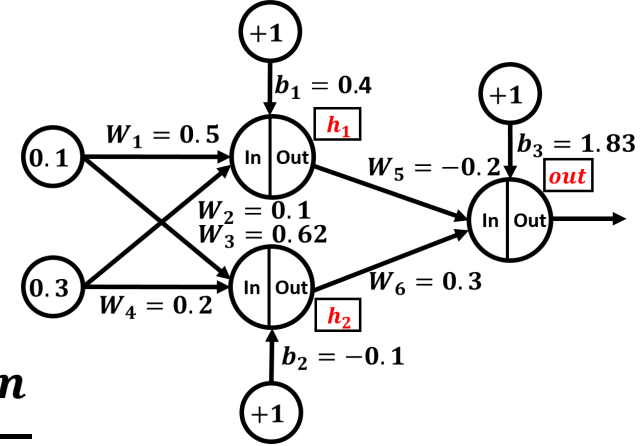
$$\begin{aligned} \frac{\partial h2_{in}}{\partial W_3} &= \frac{\partial}{\partial W_3} (X_1 * W_3 + X_2 * W_4 + b_2) \\ &= X_1 * W_3 + X_2 * W_4 + b_2 \\ &= (X_1)^{1-1} * W_3 + 0 + 0 \\ \frac{\partial h2_{in}}{\partial W_3} &= W_3 \end{aligned}$$

Substitution

$$\begin{aligned} \frac{\partial h2_{in}}{\partial W_3} &= W_3 \\ \frac{\partial h2_{in}}{\partial W_3} &= 0.62 \end{aligned}$$



$E - W_3 \left(\frac{\partial E}{\partial W_3} \right)$ Partial Derivative:



$$\frac{\partial E}{\partial W_3} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h2_{out}} * \frac{\partial h2_{out}}{\partial h2_{in}} * \frac{\partial h2_{in}}{\partial W_3}$$

$$\frac{\partial E}{\partial out_{out}} = 0.835$$

$$\frac{\partial out_{out}}{\partial out_{in}} = 0.23$$

$$\frac{\partial out_{in}}{\partial h2_{out}} = 0.3$$

$$\frac{\partial h2_{out}}{\partial h2_{in}} = 0.25$$

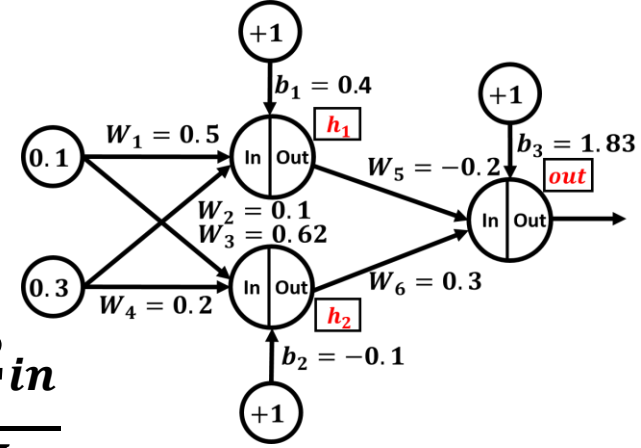
$$\frac{\partial h2_{in}}{\partial W_3} = 0.62$$

$$\frac{\partial E}{\partial W_3} = 0.835 * 0.23 * 0.3 * 0.25 * 0.62$$

$$\frac{\partial E}{\partial W_3} = 0.009$$

$E - W_4 \left(\frac{\partial E}{\partial W_4} \right)$ Partial Derivative:

$$\frac{\partial E}{\partial W_4} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h2_{out}} * \frac{\partial h2_{out}}{\partial h2_{in}} * \frac{\partial h2_{in}}{\partial W_4}$$



$E - W_4 \left(\frac{\partial E}{\partial W_4} \right)$ Partial Derivative:

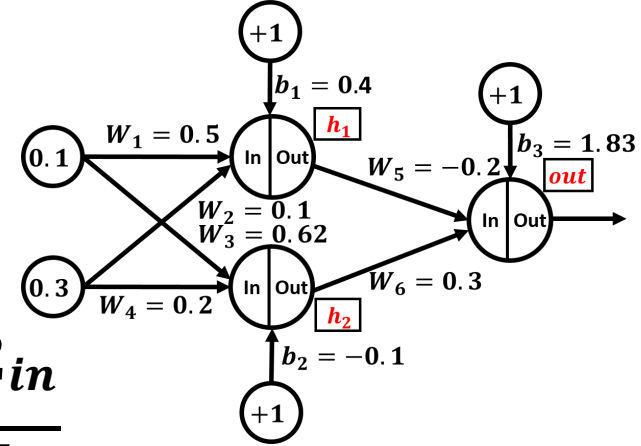
$$\frac{\partial E}{\partial W_4} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h2_{out}} * \frac{\partial h2_{out}}{\partial h2_{in}} * \frac{\partial h2_{in}}{\partial W_4}$$

$$\frac{\partial E}{\partial out_{out}} = 0.835$$

$$\frac{\partial out_{out}}{\partial out_{in}} = 0.23$$

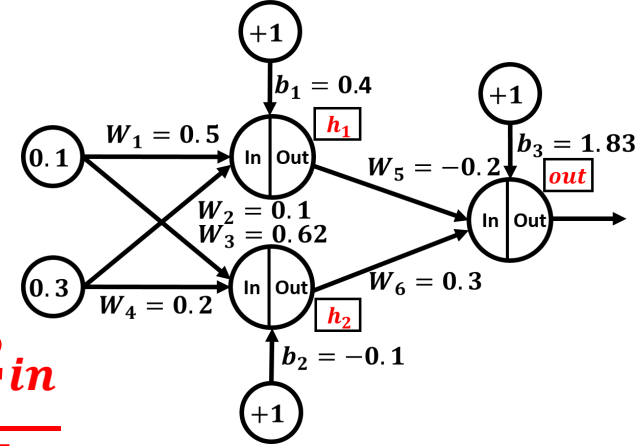
$$\frac{\partial out_{in}}{\partial h2_{out}} = 0.3$$

$$\frac{\partial h2_{out}}{\partial h2_{in}} = 0.25$$



$E - W_4 \left(\frac{\partial E}{\partial W_4} \right)$ Partial Derivative:

$$\frac{\partial E}{\partial W_4} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h2_{out}} * \frac{\partial h2_{out}}{\partial h2_{in}} * \frac{\partial h2_{in}}{\partial W_4}$$



Partial Derivative

$$\begin{aligned} \frac{\partial h2_{in}}{\partial W_4} &= \frac{\partial}{\partial W_4} (X_1 * W_3 + X_2 * W_4 + b_2) \\ &= X_1 * W_3 + X_2 * W_4 + b_2 \\ &= 0 + (X_2)^{1-1} * W_4 + 0 \end{aligned}$$

$$\frac{\partial h2_{in}}{\partial W_4} = W_4$$

Substitution

$$\frac{\partial h2_{in}}{\partial W_4} = W_4$$

$$\frac{\partial h2_{in}}{\partial W_4} = 0.2$$

$E - W_4 \left(\frac{\partial E}{\partial W_4} \right)$ Partial Derivative:

$$\frac{\partial E}{\partial W_4} = \frac{\partial E}{\partial out_{out}} * \frac{\partial out_{out}}{\partial out_{in}} * \frac{\partial out_{in}}{\partial h2_{out}} * \frac{\partial h2_{out}}{\partial h2_{in}} * \frac{\partial h2_{in}}{\partial W_4}$$

$$\frac{\partial E}{\partial out_{out}} = 0.835$$

$$\frac{\partial out_{out}}{\partial out_{in}} = 0.23$$

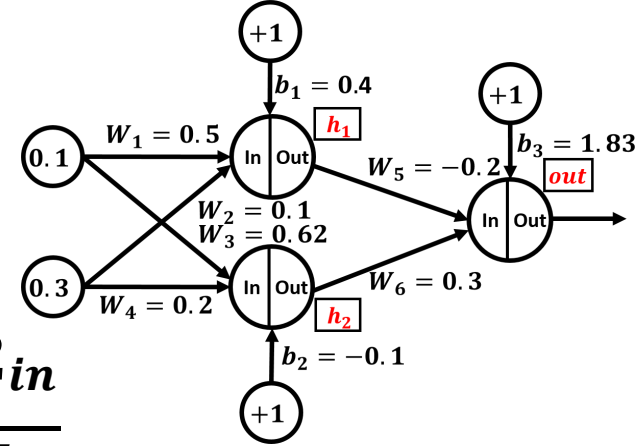
$$\frac{\partial out_{in}}{\partial h2_{out}} = 0.3$$

$$\frac{\partial h2_{out}}{\partial h2_{in}} = 0.25$$

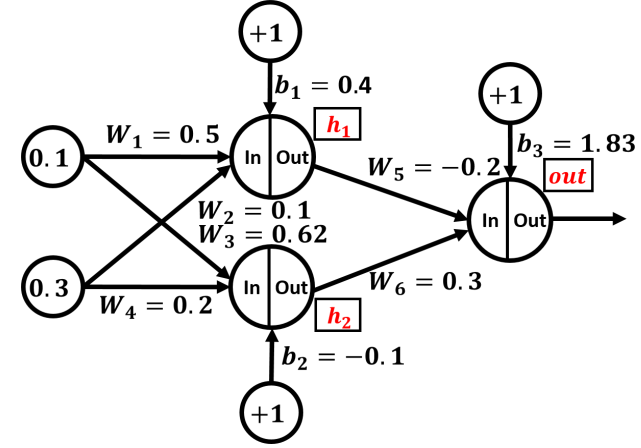
$$\frac{\partial h2_{in}}{\partial W_4} = 0.2$$

$$\frac{\partial E}{\partial W_4} = 0.835 * 0.23 * 0.3 * 0.25 * 0.2$$

$$\frac{\partial E}{\partial W_4} = 0.003$$



All Error-Weights Partial Derivatives



$$\frac{\partial E}{\partial W_1} = -0.001$$

$$\frac{\partial E}{\partial W_3} = 0.009$$

$$\frac{\partial E}{\partial W_5} = 0.119$$

$$\frac{\partial E}{\partial W_2} = -.003$$

$$\frac{\partial E}{\partial W_4} = 0.003$$

$$\frac{\partial E}{\partial W_6} = 0.097$$

Updated Weights

$$W_{1new} = W_1 - \eta * \frac{\partial E}{\partial W_1} = 0.5 - 0.01 * -0.001 = \mathbf{0.50001}$$

$$W_{2new} = W_2 - \eta * \frac{\partial E}{\partial W_2} = 0.1 - 0.01 * -0.003 = \mathbf{0.10003}$$

$$W_{3new} = W_3 - \eta * \frac{\partial E}{\partial W_3} = 0.62 - 0.01 * 0.009 = \mathbf{0.61991}$$

$$W_{4new} = W_4 - \eta * \frac{\partial E}{\partial W_4} = 0.2 - 0.01 * 0.003 = \mathbf{0.1997}$$

$$W_{5new} = W_5 - \eta * \frac{\partial E}{\partial W_5} = -0.2 - 0.01 * 0.618 = \mathbf{-0.20618}$$

$$W_{6new} = W_6 - \eta * \frac{\partial E}{\partial W_6} = 0.3 - 0.01 * 0.097 = \mathbf{0.29903}$$

Continue updating weights according to derivatives and re-train the network until reaching an acceptable error.